

BEHAV3D Tumor Profiler to map heterogeneous cancer cell behavior in the tumor microenvironment

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eLife Assessment

This is a **useful** tool for code-less analysis of patterns in cell migratory behaviours in vivo using intravital microscopy data and allows correlation with spatial features of the tumour microenvironment. There is a need for these tools to make quantitative analysis, comparison and interpretation of complex cell tracking data more accessible and evidence is provided of its applicability to tracks generated by both proprietary and open tracking software. However, it is **incomplete** due to limitations imposed by the assumptions that apply to the statistical tests employed.

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Abstract

Intravital microscopy (IVM) enables live imaging of animals at single-cell level, offering essential insights into cancer progression. This technique allows for the observation of single-cell behaviors within their natural 3D tissue environments, shedding light on how genetic and microenvironmental changes influence the complex dynamics of tumors. IVM generates highly complex datasets that often exceed the analytical capacity of traditional uni-parametric approaches, which can neglect single-cell heterogeneous in vivo behavior and limit insights into microenvironmental influences on cellular behavior. To overcome these limitations, we present BEHAV3D Tumor Profiler (BEHAV3D-TP), a computational framework that enables unbiased single-cell classification based on a range of morphological and dynamic single cell features. BEHAV3D-TP integrates with widely used 2D and 3D image processing pipelines, enabling researchers without advanced computational expertise to profile cancer and healthy cell dynamics in IVM data. Here, we apply BEHAV3D-TP to study diffuse midline glioma (DMG), a highly aggressive pediatric brain tumor characterized by invasive progression. By extending BEHAV3D-TP to incorporate tumor microenvironment

(TME) data from IVM or fixed correlative imaging, we demonstrate that distinct migratory behaviors of DMG cells are associated with specific TME components, including tumor-associated macrophages and vasculature. BEHAV3D-TP enhances the accessibility of computational tools for analyzing the complex behaviors of cancer cells and their interactions with the TME in IVM data.

Introduction

Cancer is a complex disease driven by a sequence of genetic and microenvironmental changes. As a result, tumors comprise many different cell populations that are in constant evolution¹. This tumor heterogeneity has important biological and clinical implications as it differentially promotes key characteristics of individual cancer cells, including proliferation, invasion, and drug resistance². Yet, we are far from fully understanding all the dynamic and bi-directional interactions in heterogeneous tumors, and how these interactions influence cancer cell behavior at the single-cell level. Emerging single-cell technologies and analysis tools provide a new opportunity to profile individual cells within tumors. For example, recent spatial transcriptomic and proteomic profiling provides evidence for the existence of multiple distinct microenvironmental niches within one tumor³. However, these techniques can only provide static information and make use of thin tissue sections that are only one cell layer thick. To investigate the real-time behavior of individual cells intravital microscopy (IVM) has been developed to study tumor cell behavior evolution over time but also in their three-dimensional native tissue environment.

IVM is a technique used to capture both spatial and temporal information at a single-cell level in living animals. Over the past years, it has provided many new insights into the dynamics of tumor heterogeneity and the development of therapy resistance^{4–15}. For example, intravital imaging of mouse breast cancer models revealed that only a small portion of tumor cells is motile¹⁵, and that fast- and slow-migrating cells are present in distinct regions of the tumor^{7,16}. Moreover, additional heterogeneity exists among migrating tumor cells that can, for instance, move randomly or directionally relative to other tumor cells and/or surrounding healthy tissue⁷. However, traditional IVM analyses have considered cancer cells or the tumor microenvironment (TME) as a homogeneous population, thereby ignoring heterogeneity at the single-cell level and the influence of the local microenvironment, where interactions between neighboring cells may affect cellular behavior. To address these complexities, recent studies have used unbiased approaches using morpho-kinetic parameters derived from live *in vitro*¹⁷ or *in vivo* imaging to classify cells and explore the single-cell behavioral landscape as an additional omic layer^{18–20}. In line with this concept of characterizing cellular dynamic properties for cell classification, we have previously developed an analytical platform termed BEHAV3D^{19,21} allowing to perform behavioral phenotyping of engineered T cells targeting cancer. While BEHAV3D was initially developed to analyze T cell migratory behavior under controlled *in vitro* conditions, we sought to expand its application to investigate tumor cell behaviors in IVM data, where the complexity of the TME presents distinct analytical challenges. This manuscript builds on our foundational work but represents a significant advancement by adapting the pipeline specifically for IVM datasets.

While significant efforts have been made to develop open-source segmentation and tracking tools for live imaging data, including IVM^{22–27}, fewer tools exist for the unbiased analysis of tumor dynamics. One major barrier is that implementing such analytical methods often requires substantial computational expertise, limiting accessibility for many biomedical researchers conducting IVM experiments. To bridge this gap, we present BEHAV3D Tumor Profiler (BEHAV3D-TP) by providing a robust, user-friendly tool that allows researchers to extract meaningful insights from dynamic cellular behaviors without requiring advanced programming skills. BEHAV3D-TP expands the capabilities of BEHAV3D by incorporating key features designed for *in vivo* tumor cell

dynamics (**Fig. 1** [↗](#)). The pipeline includes extensions that integrate tumor cell behavior with spatial features within the native TME, using either in situ (IVM) or fixed correlative imaging to reveal how distinct spatial niches influence tumor dynamics.

To further facilitate unbiased analysis of tumor dynamics, we developed BEHAV3D-TP as a flexible tool that supports both commercial and open-source data formats, ensuring broad compatibility across different research workflows. The pipeline supports 2D and 3D data formats generated by Imaris and by Fiji plugins such as TrackMate, MTrackJ, and ManualTracking, enabling processing of datasets from the most widely used segmentation and tracking solutions. We demonstrate the versatility and broad applicability of BEHAV3D-TP by applying it to a range of published tumor and healthy tissue IVM datasets.

Next, we focused on Diffuse Midline Glioma (DMG), a highly aggressive pediatric brain tumor characterized by invasive growth and its capacity to spread to other regions of the brain and the spinal cord ²⁸[↗](#). Our approach allowed us to observe associations between DMG cell migration behavior and specific microenvironmental factors, including tumor-associated macrophages and vasculature, by visualizing TME components either during or after IVM. In our study, we specifically focused on cancer cell migration as a specific feature of DMG, but our methodology could easily be extended to other cancer cell behaviors such as intravasation or metastatic colonization. Overall, BEHAV3D-TP provides an accessible computational approach for a better understanding of tumor-TME crosstalk and the functional implications of this communication in heterogeneous tumors.

Results

BEHAV3D Tumor Profiler: an accessible tool for mapping heterogeneity in tumor dynamics

To comprehensively map and analyze the dynamic heterogeneity of tumor cells within their TME we developed BEHAV3D-TP [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler [↗](#)]. To enhance accessibility and encourage widespread adoption among researchers studying single tumor cell dynamics in live imaging and inspired by previous work on democratizing advanced image analysis²⁶[↗](#),²⁹[↗](#),³⁰[↗](#), we designed BEHAV3D-TP as a modular Jupyter Notebook featuring an intuitive graphical user interface (GUI) (**Figure 1** [↗](#) and Supplementary Video 1). The platform runs directly in a web browser and requires only a Google account for access. BEHAV3D-TP implementation within the Google Colaboratory (Colab) framework not only ensures efficient cloud-based computation with free and scalable resources but also enhances tool accessibility by eliminating the need for local software installation, a common barrier for inexperienced users. Regardless of programming proficiency, users can navigate the computational pipeline to extract comprehensive insights into tumor dynamics (**Figure 1** [↗](#)). After time-lapse image file processing and preparation using various frameworks (**Figure 1a** [↗](#)), the central component of BEHAV3D-TP — called the *heterogeneity module* (**Figure 1b** [↗](#)) — enables researchers to uncover diverse patterns of tumor morphodynamic behavior. Additionally, we provide two extension modules — the *large-scale phenotyping module* (**Figure 1c** [↗](#)) and the *small-scale phenotyping module* (**Figure 1d** [↗](#)) — which integrate tumor cell behavior identified with the *heterogeneity module* with TME features based on different data types. The *large-scale phenotyping* module uses post-IVM fixed imaging data for tissue phenotyping, identifying extensive regions with specific TME characteristics, and assessing the distribution of distinct tumor cell behavioral patterns within these regions. The *small-scale phenotyping* module uses *in situ*-labelled TME structures from IVM data, facilitating the correlation of these TME features within the microenvironment of individual tumor cell neighborhoods. Both modules, in conjunction with the *heterogeneity module*, help to identify how different aspects of the TME influence distinct behavioral patterns of tumor cells.

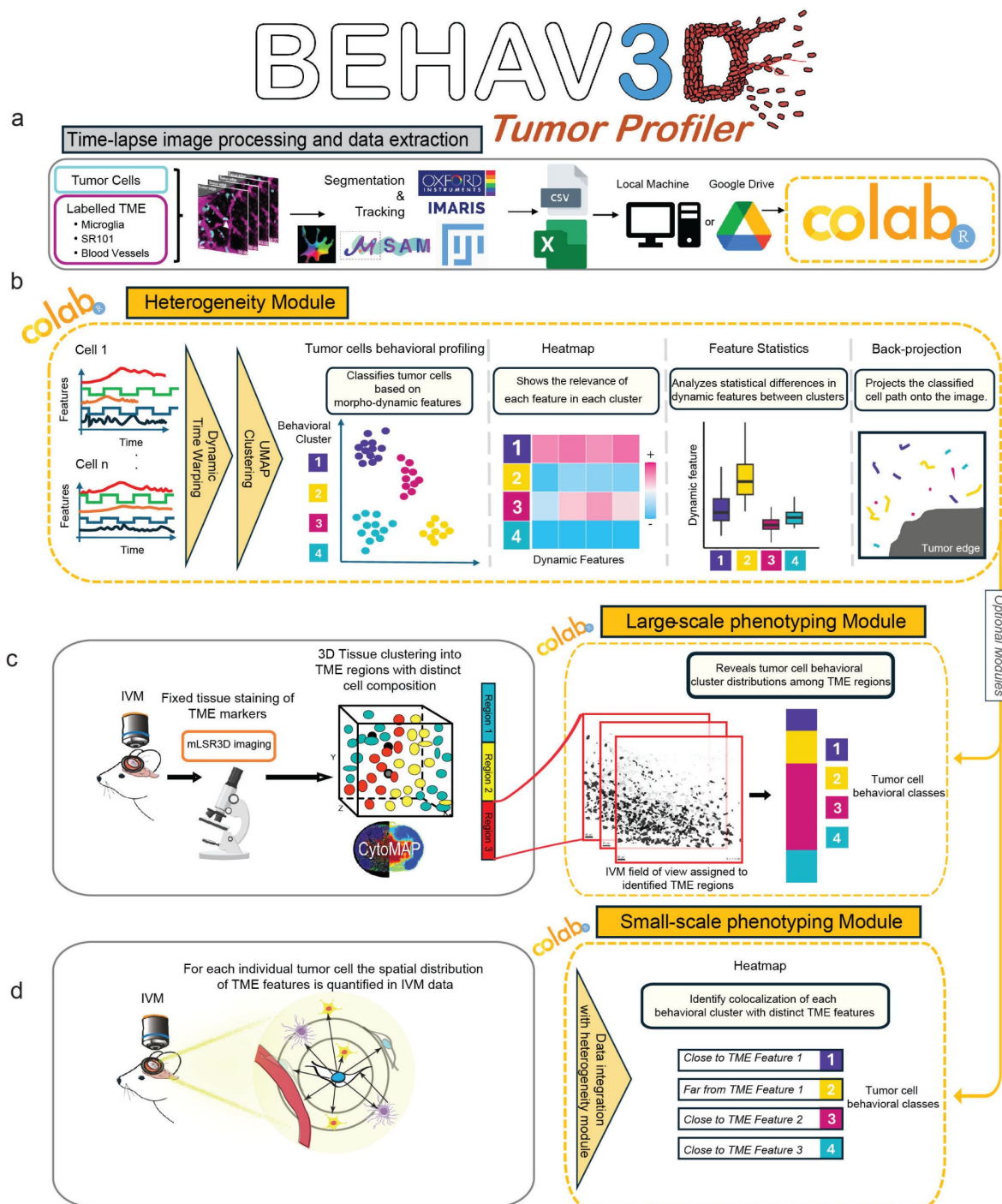


Fig. 1

BEHAV3D Tumor Profiler pipeline.

Schematic representation of the workflow showing data preparation, input and outputs in each of the three modules. **a**, Time-lapse image processing and preparation, based on segmentation and tracking of tumor (and labelled TME cells). Mounting Google Drive is optional but recommended to work in the Google Colab framework. **b**, Heterogeneity Module provides distinct behavioral clusters for cells, and relates them to dynamic features and provides a back-projection of the tracked cells. **c**, **d**, Optional modules, additional to the Heterogeneity module. **c**, Large-scale phenotyping module combines TME large-scale information with cell morphodynamic profiling to further depict population frequencies in distinct TME regions. **d**, Small-scale phenotyping module combines TME small-scale information with morphodynamic profiling to depict how the cellular environment can influence tumor cell behavior.

To evaluate the capability of the BEHAV3D-TP *heterogeneity module* in identifying diverse morphodynamic cell patterns, we applied it to IVM datasets from various tumor and healthy models. The datasets, processed with different segmentation and tracking tools, were analyzed for multivariate similarities in morphological and dynamic features (summarized in Table 1) using the dynamic time warping algorithm²¹.

As an initial benchmark, we analyzed a previously published IVM dataset of two migratory and metastatic breast cancer cell lines (4T1 and MDA-MB231), tracked in 2D using the MTrackJ Fiji plugin (Supplementary Figure 1a)¹⁵. BEHAV3D-TP provided a more granular characterization of these behaviors, uncovering additional migratory patterns not reported in the original study (Supplementary Figure 1b-c). Specifically, we identified behaviors such as *Fast*, *Intermediate*, *Very slow* and *Static* that had a different distribution among both tumor cell lines (Supplementary Figure 1d-e).

To further assess the versatility of BEHAV3D-TP, we applied it to 2D IVM data of terminal end buds (TEB) from healthy mammary gland tissue³¹. This dataset features confetti-labeled epithelial cell types and was analyzed using maximum intensity projections of multiple z planes showing cell morphology changes. As described in the original publication³¹, cell segmentation was performed with Cellpose 2.0³² within the MaSCOT-AI workflow, followed by tracking using the TrackMate plugin in Fiji. This workflow not only confirmed the compatibility of BEHAV3D-TP with advanced segmentation frameworks but also its usage to unravel distinct morphodynamic patterns in non-tumor epithelial cells (Supplementary Figure 2b,c). Consistent with previous findings that HR+ TEB (Pr-cre/Confetti) exhibit greater migratory capacity than HR-TEB (Elf5-rtTA/TetO-Cre/Confetti), our analysis suggests that this difference is primarily driven by a distinct morphodynamic behavior (Supplementary Figure 2 c,d). Specifically, HR+ TEB more frequently exhibited characteristics of Cluster 2, defined by a rounder shape and increased motility. Despite the small sample size requiring further validation, these results demonstrate BEHAV3D-TP's ability to uncover complex, multiparametric cellular behaviors that are often missed in analyses focused on single-value comparisons across conditions.

We next tested BEHAV3D-TP with a 3D image processing pipeline applied to IVM data from adult glioma (GBM)⁷. In this case, cells were segmented using μ SAM³³, a deep-learning foundation model adapted for microscopy data (Segment Anything for Microscopy), and subsequently tracked with TrackMate²⁷ Fiji plugin (Supplementary Figure 3a). BEHAV3D-TP identified distinct GBM morphodynamic patterns, each characterized by unique combinations of morphological and dynamic features and associated with specific spatial distributions (Supplementary Figure 3b-d). Cluster 1, consisting of large and slow-moving cells, was more isolated and located farther from neighboring cells compared to Cluster 6 (small, fast, persistent cells) and Cluster 4 (very small, static cells). These findings suggest that the compact spatial arrangement of small cells within the tumor may contribute to their reduced size.

Collectively, these findings underscore the versatility and broad applicability of BEHAV3D-TP across diverse biological systems, dimensionalities (2D and 3D), imaging modalities, and data processing pipelines, supporting its utility as a robust tool for the in-depth analysis of cellular behavioral patterns in IVM data.

Identification of highly heterogeneous *in vivo* cell behavior displayed by DMGs

We then applied our platform to get insights into the behavioral heterogeneity among pediatric DMG tumors, characterized by a highly invasive nature. We implanted mice with patient derived DMG cells expressing H2BmNeonGreen marker and performed IVM for up to 5.4 hours upon tumor development (Figure supplement 4a). We tracked intravitaly imaged DMG cells from different mice using Imaris, a commercial software widely adapted by the IVM community³⁴.

⁴⁰ for its intuitive 3D visualization and analysis capabilities. In each imaged position, we evaluated not only the dynamic characteristics of individual DMG cells but also their spatial relationship to the tumor edge, which was delineated using Imaris Surface module (**Figure 2a**, see Methods). Next, to untangle the complexity of tumor cell dynamics in an unbiased manner, we used the BEHAV3D-TP *heterogeneity module*, implementing multiparametric single-cell time-series classification to identify distinct single-cell behavioral patterns (**Figure 2 a, b**). For this, we used kinetic parameters, including displacement, speed, persistence, and movement direction relative to the tumor edge (Table 1), followed by dimensionality reduction to classify cells. The BEHAV3D-TP *heterogeneity module* identified 7 DMG behavioral clusters: *retreating*, *slow retreating*, *erratic*, *static*, *slow*, *slow invading*, and *invading* (**Figure 2b, c, d**, Supplementary video 2). To accurately describe each of the clusters, we analyzed their most predominant features (**Figure 2d**) and back-projected the cluster information into the original time-lapse to visually inspect cell behavior in relation to the tumor (**Figure 2a, c**, Figure supplement 4b, Supplementary video 2). Of particular interest were cells from the *invading* and *retreating* clusters, characterized by fast movement away from and towards the tumor edge, respectively (**Figure 2c-f**, and Figure supplement 4c). In fact, in this particular DMG model, approximately 10% of the cancer cells were displaying *invasive* behavior (Figure supplement 4d), in line with the intrinsic infiltrative nature of DMG⁴¹. Interestingly, about another 10% of DMG cells were moving back to the tumor edge (*retreating*, *slow retreating* clusters) (Figure supplement 4d), behavior that has been previously observed for glioblastoma and suggested to be regulated by differential chemotactic signaling by the tumor edge and the surrounding brain parenchyma⁷. We also observed *erratic* cells that display fast non-directed migration behavior, and *static* cells that are non-migratory (**Figure 2c-f**). Crucially, the distinct behavioral patterns were detected in all intravitaly imaged mice (Figure supplement 4d), ruling out any sample bias that might lead to clusters consisting solely of cells from a single tumor or mouse. This underscores the effectiveness of the BEHAV3D-TP in revealing representative heterogeneous tumor behavioral patterns.

Mapping *in vivo* tumor cell migratory properties to distinct TME regions identified through *ex vivo* large-scale 3D imaging

To investigate whether the TME influences behavioral heterogeneity among cancer cells, we developed two complementary approaches to correlate the identified behavioral clusters (**Figure 2**) to the composition of the TME (**Figures 1c**, **3a** and **4a**).

The first approach consists of performing large-scale TME phenotyping and identifying regions with a specific cellular composition and architecture within the TME of intravitaly imaged tumors that were subsequently fixed. This is accomplished by using multispectral large-scale single-cell resolution 3D (mLSR-3D) imaging data^{42,43} and a spatial analysis framework called CytoMAP⁴⁴ (**Figure 3a**). We selected microenvironmental components of interest, such as perivascular niches and glial cells, as they have been previously reported to play an important role in the invasive behavior of glioblastoma cells^{7,45–49}. Specifically, intravitaly imaged brains were collected immediately after the IVM imaging sessions and were fixed and stained for blood vessels (CD31), oligodendrocytes (Olig2), and tumor-associated microglia/macrophages (TAMM) (Iba1) (**Figure 3b**). Subsequent CytoMAP⁴⁴ spatial phenotyping analysis identified three different environmental niches: *Void* regions (less vascularized and with low glial infiltration), *TAMM/oligo* regions (enriched in oligodendrocytes, TAMM, and tumor cells), and *TAMM/vascularized* regions (highly vascularized regions enriched in TAMM) (**Figure 3b**, and Figure supplement 5a-c). To investigate the heterogeneity of tumor cell behavior in relation to the identified microenvironmental niches, BEHAV3D-TP *large-scale phenotyping module* mapped the different niches (*Void*, *TAMM/Oligo*, and *TAMM/vascularized*) identified by CytoMAP onto 3D intravital imaging data (**Figures 1c**, **3c**). In the TME-defined IVM-imaged positions we compared the frequencies of the different behavioral clusters that we identified with the BEHAV3D-TP *heterogeneity module* (**Figure 3d**). Interestingly, *invading* cells were more

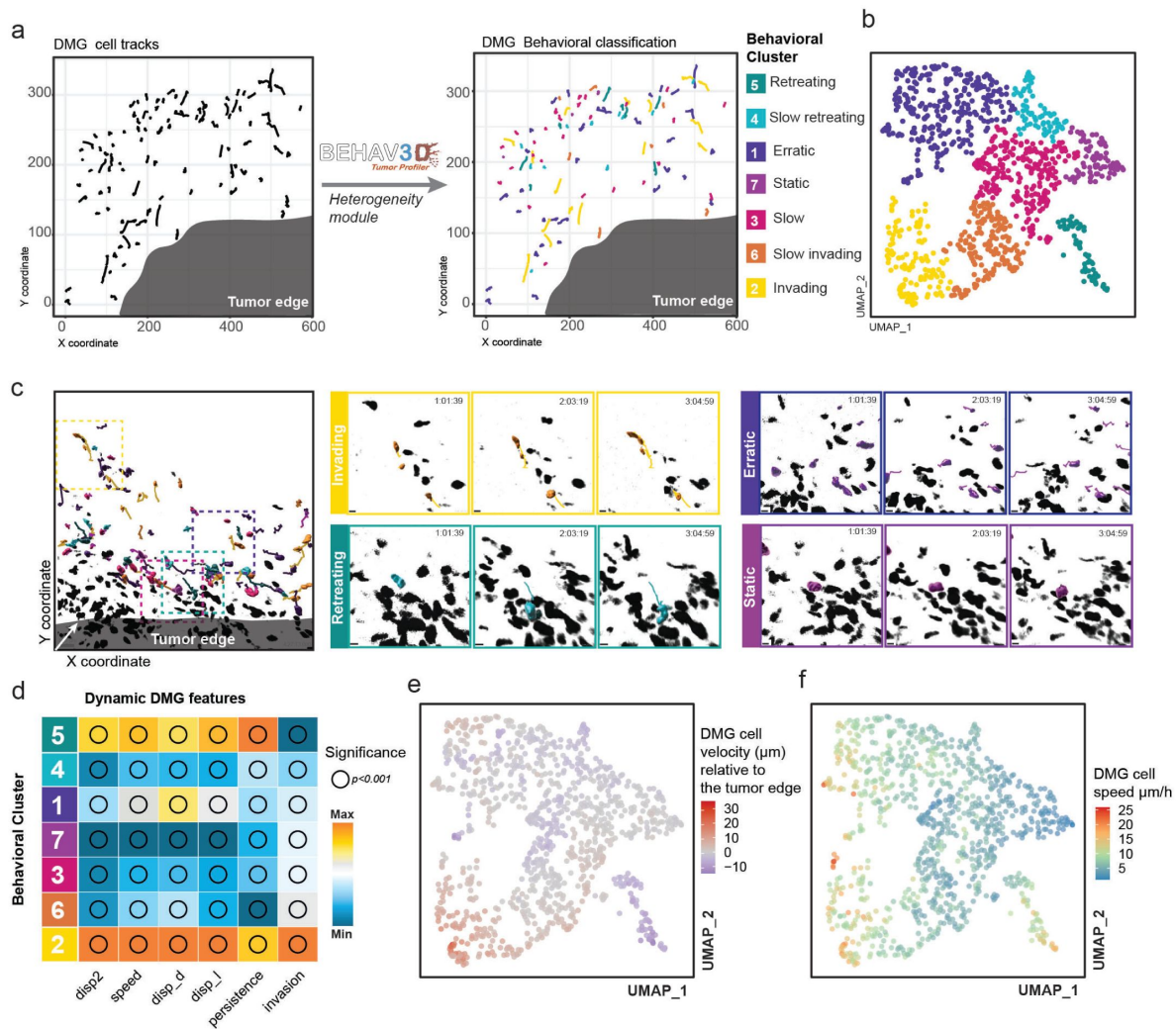


Fig. 2

Behavioral profiling of DMG cells with BEHAV3D-TP heterogeneity module.

a, Representative intravitaly imaged position showing DMG cells tracked relative to tumor Edge (left panel). Tracks were classified according to DMG behavior (right, color coded by cluster). Representative of $n = 18$ independent positions from $n=6$ mice. **b**, UMAP plot showing seven color-coded DMG cell behavioral clusters identified by BEHAV3D-TP. Each datapoint represents one DMG cell track. A total of 981 individual cells were tracked in $n = 18$ independent positions from $n=6$ mice. **c**, Representative intravitaly imaged positions and enlarged sections for showcasing distinct DMG behavioral clusters *Invading* (yellow), *Retreating* (green), *Erratic* (dark purple) and *Static* (light purple) through the time series. $n = 18$ independent positions from $n=6$ mice. Scale bars 10µm. **d**, Heatmap depicting relative values of DMG cell features indicated for each cluster, named according to their most distinct characteristics. Arbitrary units in respect to maximal and minimal values for each feature. See Table 1 for features description. Bubble size according to significance levels, all features represented have a significance level of $P < 0.001$ (***): *mean_displ2*, *mean_squared displacement*; *speed*; *delta displacement*; *displacement length*; *persistence*; *invasion*. P-values represent the significance of differences across clusters tested through ANOVA for each feature. **e**, **f**, UMAP representation of DMG cells' velocity relative to the tumor edge (µm) (**e**) and speed (µm/h) (**f**). Each datapoint represents one cell track and is colored following a color gradient. For each feature colors represent the mean DMG track values. Representative of $n = 18$ independent positions from $n=6$ mice.

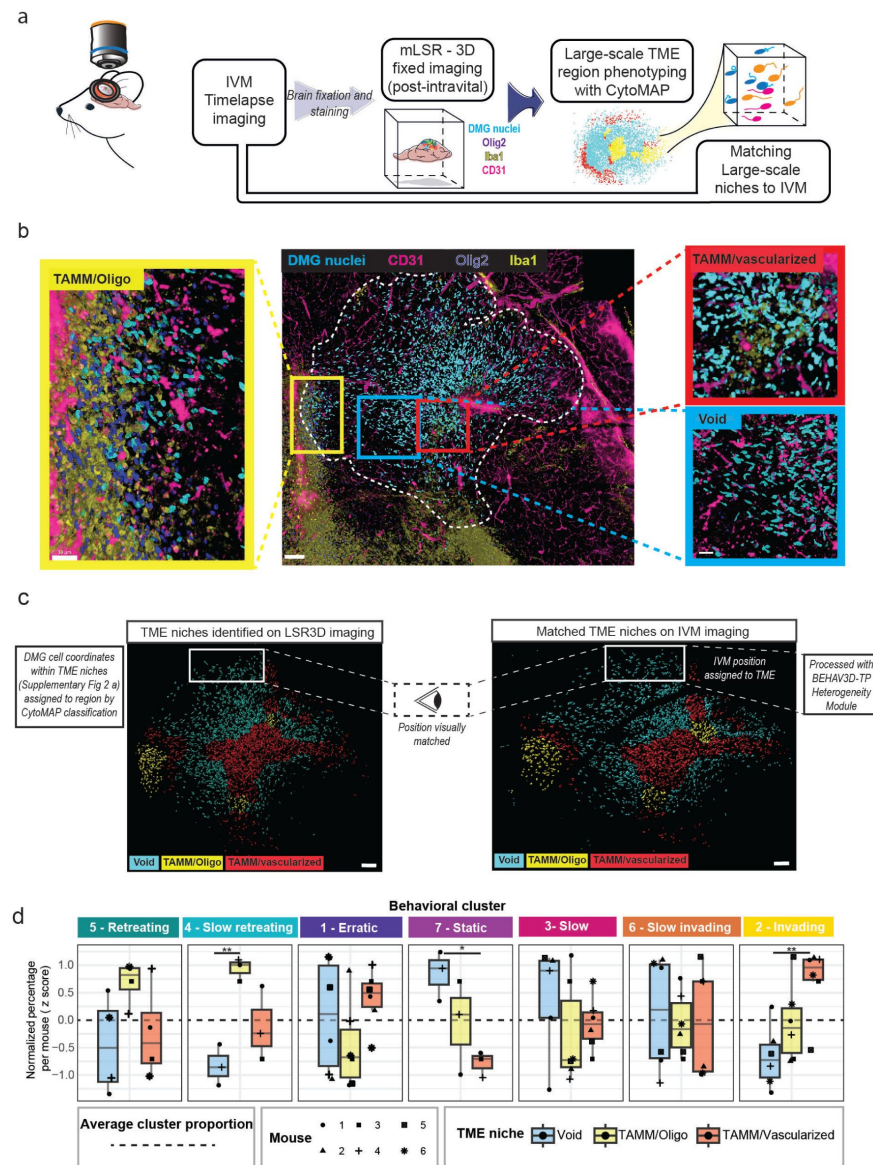


Fig. 3

Mapping distinct DMG behavioral patterns to distinct TME regions.

a, Overview of the large-scale TME region phenotyping workflow. After IVM imaging session, the tissue is fixed and stained and then analyzed using mLSR-3D. CytoMAP Spatial Analysis Toolbox is used to identify distinct large-scale regions, that are subsequently matched to IVM imaging dataset (**c**). **b**, Representative fixed multispectral image of a mLSR3D imaged DMG tumor and zoomed-in images of TME large-scale regions: *Void* (blue outline), *TAMM/vascularized* (red outline) and *TAMM/Oligo* (yellow outline). DMG nuclei (blue), CD31 (pink), Olig2 (purple), and Iba1 (yellow) are represented on the imaged. Scale bars, 100 μ m (overview), 30 μ m (zoomed-in regions). **c**, 3D multispectral image DMG cell rendering color-coded per large TME region: *Void* (blue), *TAMM/Oligo* (yellow) and *TAMM/vascularized* (red) (left). Based on the spatial coordinates of DMG cells classified into specific TME regions by CytoMAP, cells within the mLSR-3D images were subsequently assigned to their respective TME regions. Regions were then visually mapped on intravital imaging data (right). Representative of $n=6$ mice. White outline indicates one of the tumor regions analyzed with BEHAV3D-TP from the full tumor volume. **d** Boxplots showing the frequencies of different DMG behavioral clusters in *Void*, *TAMM/Oligo* and *TAMM/vascularized* TME large-scale regions. For each behavioral cluster, the y-axis displays the z-scored percentage of cells, normalized per mouse to account for inter-mouse variability. Each point represents an individual imaged position, with shape indicating the mouse of origin. P-values represent the significance of differences across TME regions tested through ANOVA with Tukey post hoc, for each behavioral cluster. $n = 18$ independent positions from $n=6$ mice.

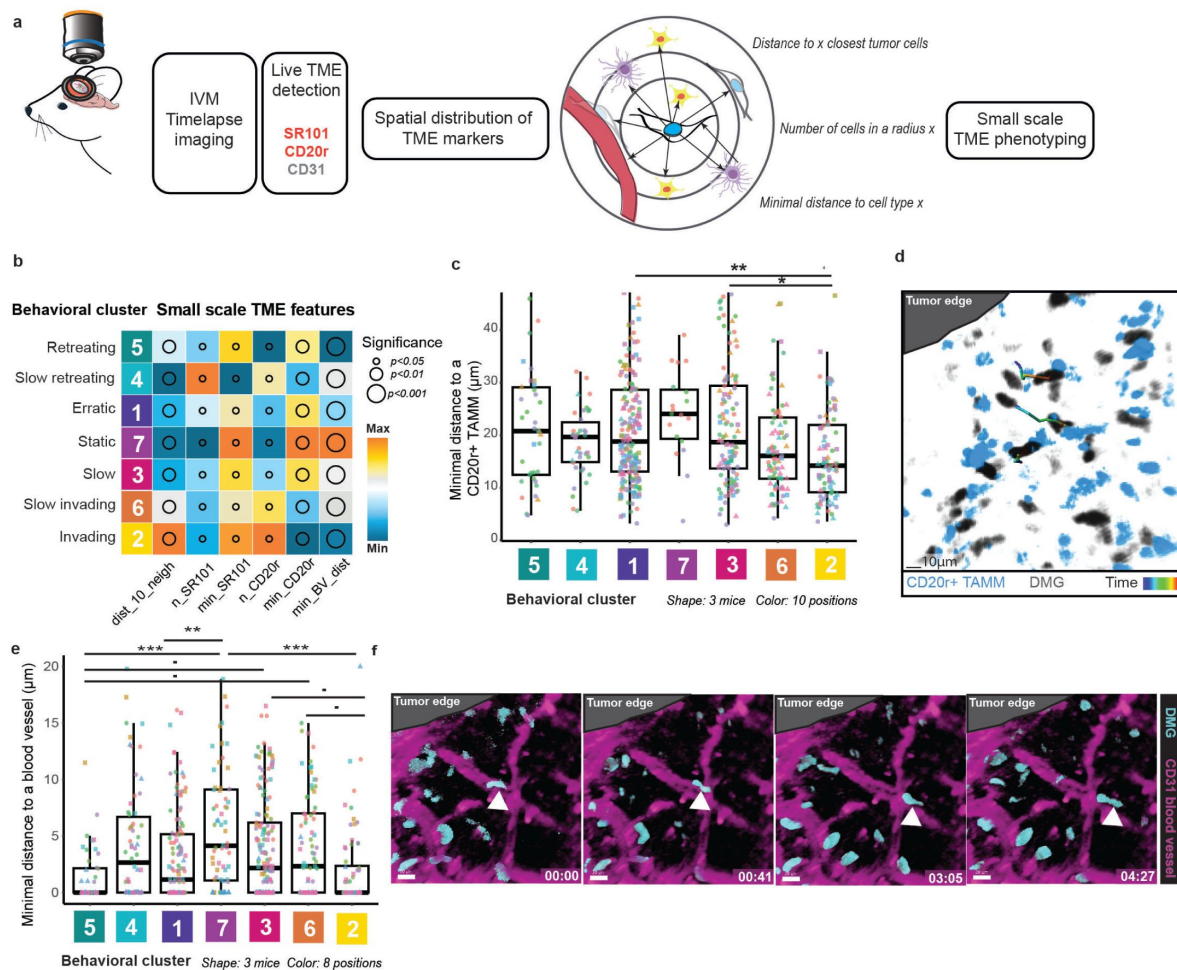


Fig. 4

The BEHAV3D-TP small-scale phenotyping correlates DMG behavioral profiles with microenvironmental composition at a single-cell resolution.

a, Overview of the small-scale TME phenotyping module. During IVM imaging, information about spatial distribution of SR101+, CD20r+, and CD31+ cells is collected and processed by the BEHAV3D-TP *small-scale phenotyping module*. Single DMG cell spatial TME features like distance to its neighbors, number of cells in a radius or minimal distance to a cell type were measured. **b**, Heatmap depicting DMG cell small-scale niches' features across distinct DMG behavioral clusters. Arbitrary units in respect to maximal and minimal values for each feature. Bubble size according to significance levels, from smallest ($P < 0.05$ (**)) to medium ($P < 0.01$ (**)) to biggest ($P < 0.001$ (***)). *dist_10_cell*, Distance to nearest 10 cells, $P < 0.01$ (**); *n_SR101*, number of SR101 cells, $P < 0.05$ (*); *min_SR101*, minimal distance to a SR101 cell, $P < 0.05$ (*); *n_CD20r*, number of CD20r cells, $P < 0.05$ (*); *min_CD20r*, minimal distance to a CD20r cell, $P < 0.01$ (**); *min_BV_dist*, minimal distance to a Blood vessel, $P < 0.001$ (***)). P-values represent the significance of differences across clusters tested through ANOVA for each feature. **c**, **e**, Boxplot depicting the *Minimal distance to a CD20r+ TAMM* (μm) (**c**) and the *Minimal distance to a blood vessel* (μm) (**e**) across DMG behavioral clusters. Each point represents an individual tracked cell, color depicts the imaged position and shape denotes the mouse. P-values represent the significance of differences across behavioral clusters tested through ANOVA with Tukey's post hoc. $P < 0.01$ (**); $P < 0.05$ (*); $P < 0.001$ (***) ; $P < 0.1$ (.). **c**, $n = 10$ independent positions from $n=3$ mice. **e**, $n = 8$ independent positions from $n=3$ mice. **d**, Multispectral image showing the movement of DMG cells (grey) relative to CD20r+ TAMM (blue) and the Tumor core. The trajectories show a path from the initial (blue) to the final position (red). Scale bar, 10 μm . **f**, Time lapse multispectral images (left to right) showing the DMG cells (blue) displacement along CD31 blood vessel (purple) away from the tumor edge. Scale bars, 30 μm .

abundant in *TAMM/vascularized* regions compared to *Void* regions that contain a higher proportion of *static* cells, and *TAMM/Oligo* microenvironments contain more *slow retreating* cells compared to the other regions (Figure 3d, Figure supplement 5d). Finally, we compared the results obtained with BEHAV3D-TP to a more classical IVM analysis approach that relies on assessing single dynamic parameters. Interestingly, single parameters showed a restricted ability to identify differences in cellular behavior among various environments (Figure supplement 5e-g), underscoring the importance of multiparametric behavioral mapping in revealing more nuanced cellular behavior that cannot be captured solely by individual dynamic parameters.

Linkage of *in vivo* tumor cell behavior to the composition of the microenvironment at a single-cell level

With the goal of further refining TME phenotyping to better understand tumor cell behavior determinants, we implemented an alternative approach using the BEHAV3D-TP *small-scale phenotyping module* (Figure 1d, 4a). This module utilizes the detection of TME components during IVM to offer insights into their abundance and spatial relationship with DMG cells at a single-cell level. Together with the *heterogeneity module* that profiles the single-cell behavior of tumor cells (Figure 1c), this additional module allows linking different DMG behavioral profiles with DMG cell small-scale niches (Figures 1d, 4a). For *in vivo* TME labeling, we injected mice with an anti-CD31-AF647 antibody to label blood vessels, and with either SR101, a fluorescent marker known to label oligodendrocytes and astrocytes^{50–53}, or CD20r, a novel molecule described to label microglial cells⁵⁴ (Figure supplement 6a, b). This allowed us to measure spatial TME characteristics of each single DMG cell, such as minimal distance to a certain TME component, the number of these cell types in a certain radius, or the cell density for this particular cell (Figure 1d). Interestingly, the BEHAV3D-TP *small-scale phenotyping module* revealed substantial TME heterogeneity across the different behavioral DMG clusters (Figure 4b). Compared to *slow* and *slow-retreating* cells, *invading* cells were usually found to be less densely positioned relative to other DMG cells (Figure 4b, and Figure supplement 6c), correlating with their invasive potential into the brain parenchyma. In line with the data from the large-scale TME phenotyping (Figure 3c), our data suggests that *invading* DMG cells reside in a local environment that is enriched in TAMMs (Figure 4b). Indeed, *invading* cells are closer to TAMMs compared to *erratic* and *slow migrating* cells (Figure 4c), and move directionally towards TAMMs (Figure 4d, Supplementary video 3), suggesting that migration of this behavioral cluster is at least partially mediated by chemotaxis. Furthermore, all fast-migrating behavioral clusters (*invading*, *erratic*, and *retreating*) were found to be generally closer to blood vessels (Figure 4b and e), suggesting that perivascular invasion is a more efficient movement route for these cells (Figure 4f), a mode of migration that has been previously observed for adult glioblastoma cells as well^{7,45,47–49}. Lastly, particularly *slow-retreating* DMG cells were found in regions enriched for SR101⁺ cells (Figure 4b and Figure supplement 6d), perhaps reflecting oligodendrocyte-like (OC-like) tumor DMG cells that have been found near oligodendrocytes⁵⁵. Altogether, these data demonstrate the power of our BEHAV3D-TP pipeline in correlating single-cell tumor cell behavior to its local environment, with an important opportunity to identify novel environmental regulators of invasion.

Discussion

Here, we provide a new user-friendly platform tailored for IVM single tumor cell dynamics analysis, termed BEHAV3D-TP, aiming to better understand the mechanisms underlying tumor heterogeneity. In contrast to traditional pipelines that rely on the analysis of single parameters such as cell speed or displacement^{56–58} or that use an artificial threshold to assign a specific behavior to cells¹⁰, BEHAV3D-TP is an unbiased approach based on the integration of multiple parameters measured by IVM. Using the *heterogeneity module* of our BEHAV3D-TP pipeline, we uncovered behavioral phenotypes that have not been described before and that could not have

been identified using single parameter analysis. For example, our analysis revealed *invading* and *retreating* clusters, composed of cells with a similar cell speed but with an opposite migration direction, an important biological and clinical difference. Likely, bidirectional migration is mediated by different chemotactic cues and by the composition of the local TME⁵⁹. It is of vital importance for DMG and other cancer patients to better understand the biology underlying the different behavioral phenotypes, especially concerning *invading* cells that are difficult to therapeutically target⁶⁰.

Previous studies suggest a major role for the microenvironment in promoting tumor cell spread into the healthy brain^{61–63}. Indeed, using the complementary *large-* and *small-scale TME phenotyping modules* of BEHAV3D-TP, we identified substantial differences in the TME composition of the different behavioral DMG clusters. Our results also show that the small-scale TME analysis has a higher predictive value compared to the large-scale TME analysis, suggesting that a cell's direct microenvironment is a more important regulator of its behavior compared to its macroenvironment. Future experiments directed towards identifying more TME components, for example using multispectral LSR-3D imaging⁴², could therefore further refine the *large-scale phenotyping module* by detecting subregion heterogeneity.

IVM enables simultaneous labeling of various TME components and allows real-time investigation of their association with tumor cell behavior. Indeed, we found that TAMMs and blood vessels were associated with specific behavioral phenotypes, showing the power of BEHAV3D-TP *small-scale phenotyping module* in identifying novel regulators of cellular behavior. This is exemplified by our finding that *slow-retreating* DMG cells were found in environments enriched for SR101⁺ cells, an observation that has not been made before. To take further advantage of BEHAV3D-TP, future work should be directed toward identifying more TME components, including cells of the adaptive immune system. For example, it would be interesting to use our pipeline on recently developed immunocompetent mouse models of DMG^{64,65}, or in mice with a humanized immune system⁶⁶. While our findings suggest that microenvironmental factors may influence tumor cell migration, further studies will be necessary to establish causal relationships. Additional experimental validation, such as macrophage ablation experiments, could help clarify the specific contributions of these factors.

While our current study does not provide direct functional validation of the distinct motility clusters identified, existing literature strongly supports the notion that cell dynamics can serve as a proxy for functional states and phenotypic heterogeneity. Prior work, including studies by our group^{19,67} as well as Crainiciuc et al.³⁶ and Freckmann et al.²⁰, has demonstrated that variations in cell motility patterns can reflect underlying functional characteristics. Specifically, cell morpho-dynamic features have been shown to correlate with differences in cell type identity, T-cell engagement, metastatic potential, and drug resistance states. This growing body of evidence suggests that tumor cell behavior, as captured by BEHAV3D-TP, may serve as a predictive tool for deciphering functional tumor heterogeneity. Future studies integrating transcriptomic or proteomic profiling of motility-defined subpopulations could further elucidate the biological significance of these behavioral phenotypes.

In addition to motility-based classification, features such as tumor cell morphology, proliferation state, and interactions with the tumor microenvironment can further refine tumor phenotyping. BEHAV3D-TP allows for the selection of diverse feature types, supporting datasets that include both dynamic, morphological and spatial parameters. However, we recognize that expanding the feature set may introduce biologically irrelevant noise, particularly in 3D microscopy data where limited z-axis resolution can lead to morphological artifacts. This highlights the potential need in the future to include unbiased feature selection strategies, such as bootstrapping methods⁶⁸, to ensure the identification of meaningful and biologically relevant parameters. Careful consideration of these aspects is key to maximizing the interpretability and predictive value of analyses performed with BEHAV3D-TP.

To enhance the accessibility of BEHAV3D-TP for researchers keen on studying tumor cell behavioral profiles through live imaging, we have incorporated our software in a Google Colab notebook, allowing researchers with minimal programming skills to employ the tool online without the requirement to install specific software environments nor the need to directly interact with the code. The user can easily install the necessary dependencies, set up the working environment, process datasets and adjust parameters without requiring deep coding skills. Google Colab provides the necessary computational resources to be able to run the pipeline without demanding a powerful local machine. However, there are RAM usage and runtime limitations that could affect the analysis, given a big enough dataset in the free version of Google Colab. Given this and data privacy concerns, the workflow can also be run locally via a Jupyter Notebook, that can be downloaded from the Google Colab notebook. Aligned with the goals of BEHAV3D-TP, other methods utilizing the Google Colab framework are emerging, such as CellTracksColab²⁹, which is a general tool for tracking data analysis. While CellTracksColab offers valuable insights into tracking data, BEHAV3D-TP is specifically designed for IVM data and includes additional features. These features include directionality metrics crucial for invasion studies, detailed analysis of identified behavioral populations, data back-projection, and correlation of these behaviors with both large-scale and small-scale TME features. This comprehensive approach provides a deeper understanding of the TME drivers behind intratumoral heterogeneity, making BEHAV3D-TP a robust tool for researchers investigating the complex behaviors of tumor cells in their natural environments. BEHAV3D-TP supports a broad range of data formats, including those generated by commercial Imaris and by open-source Fiji plugins such as TrackMate, MTrackJ, and ManualTracking, increasing the tool's applicability to researchers using both commercial and open-source pipelines.

In summary, BEHAV3D-TP allows to get insights into the heterogeneity of tumor cell migration behavior in an accessible way; however, it would be exciting to extend the application to other behaviors, such as cell division or therapy response to study TME-mediated drug resistance.

Methods

Animal material

NOD.Cg-PrkdcscidIl2rgtm1Wjl/SzJ (NSG) mice were purchased from Charles River Laboratories (France). Experiments were conducted per Institutional Guidelines under acquired permission from the local Ethical Committee and as per current Dutch laws on Animal Experimentation. Mice were housed under sterile conditions using an individually ventilated cage (IVC) system and were fed with sterile food and water.

Human DMG sphere-forming culture

DMG007 (HSJD-DIPG-007) cells established from primary DMG material were grown in a base medium (TSM) consisting of 48% DMEM/F12 and 48% Neurobasal medium (Thermo Fisher) supplemented with GlutaMAX, 100mM Sodium Pyruvate, MEM non-essential amino acids and 1M Hepes buffer, at 1% each. Primocin (InvivoGen) at 50mg/ml was additionally added to the TSM. Working medium was prepared by adding 20ng/ml of human EGF and human basic FGF, 10ng/ml of PDGF-AA and PDGF-BB (Peprotech) and 2ug/ml of Heparin (StemCell Technologies). DMG cells were cultured at 37 °C in a humidified atmosphere with 5% CO₂.

Fluorescent reporter cloning and lentiviral transduction

Fluorescent reporter construct was based on the combination of Histone2B-mNeonGreen, P2A and mScarlet I-CAAX gifted from Dr. Hugo Snippert (UMC Utrecht, The Netherlands), and inserted by In-Fusion HD Cloning Plus (TakaraBio) into a lentiviral vector including cPPT/CTS, WPRE, an EF1a promoter and IRES-puromycin-resistance cassette¹¹. Subsequently the above-mentioned

elements were amplified by PCR and the amplicon was assembled in the final lentiviral backbone using Gibson Assembly (NEB). Sequences were validated using Sanger sequencing. DMG007 cells were transduced using a standard lentiviral transduction protocol and selected using puromycin to achieve a stable cell line.

Cranial imaging window (CIW) surgery and tumor cell injection

CIW surgery and tumor cell injection were performed on the same day as previously described [7,10](#). Briefly, mice were sedated with Hypnorm (Fluanison [neuroleptic] + Fentanyl [opioid]) (0.4 ml/kg) + Midazolam [benzodiazepine sedative] (2 mg/kg) at a dose of 1:1:2 in sterile water and mounted in a stereotactic frame. The head was secured using a nose clamp and two custom-made 3D printed ear bars, made out of Polylactic Acid (PLA, Geeetech, 700-001-043). The head was shaved, and the skin was cut circularly. The mouse was placed under a stereo-microscope to ensure precise surgical manipulation. The periosteum was scraped, and a circular groove of 5 mm diameter was drilled over the right parietal bone. The bone flap was lifted under a drop of cortex buffer (125 mM NaCl, 5 mM KCl, 10 mM glucose, 10 mM HEPES buffer, 2 mM MgSO₄, and 2 mM CaCl₂, pH 7.4), and the dura mater was removed. Gelfoam sponge (Pfizer) was used to stop potential bleedings. Next, 1×10^5 DMG007-H2B-mNeonGreen cells resuspended in 3 μ l phosphate-buffered saline (PBS) were injected stereotactically using a 5 μ l Hamilton syringe with a 2 pt style in the middle of the craniotomy at a depth of 0.5 mm. The exposed brain was sealed with silicone oil and a 6 mm coverslip was glued on top. Dental acrylic cement (Vertex) was applied on the skull surface, covering the edge of the coverslip, and a custom-made 3D printed ring made of biocompatible Polylactide acid (PLA, Geeetech, 700-001-0433) was glued around the coverslip to provide fixation during intravital imaging and to serve as a reservoir for a water drop required for the objective of the microscope. A single dose of 100 μ g/kg of buprenorphine (Temgesic, BD pharmaceutical limited) was administered for post-surgical pain relief. Mice were closely monitored twice per week to assess behavior, reactivity, and appearance.

Intravital imaging

For DMG IVM imaging mice were intravitaly imaged as previously described [7,10](#). In short, mice were sedated with isoflurane and placed face-down in a stereotactic frame. The Polylactide ring was used to fix the mouse's head to the frame with custom-made Polylactide bars. Time-lapse images of the entire tumor volume were acquired for a maximum of 5.4 hours. The minimal time interval between serial images was set to 20 minutes. For tile scans, images of the complete z-stack of the tumor were acquired, with a step size of 2 μ m. In a group of 3 mice, oligodendrocytes and astrocytes were imaged by an intravenous injection of 50 μ L SR101 (ThermoFisher) at a concentration of 5 mM dissolved in PBS. In a different group of 3 mice, TAMMs were imaged by intravenous injection of 50 μ L 100 μ M CDr20 [54](#). Simultaneously with the injection of either SR101 or CDr20, all mice received an intravenous injection of 10 μ L CD31-AF647 antibody (ThermoFisher, A14716) to visualize the blood vessels.

Intravital imaging was performed on an upright FVMPE-RS multiphoton microscope (Olympus) equipped with a MaiTai DeepSee and an InSight DeepSee laser (Spectra-Physics), a 25x/1.05 numerical aperture water objective, and two GaAsP and two Multialkali photomultiplier tubes (Olympus). Images were acquired with simultaneous imaging with the MaiTai (960 nm) and InSight (1100 nm) lasers. Green, red, and far-red fluorescence was separated by 570 nm and 650 nm dichroic mirrors, and 610/70 nm and 705/90 nm filters (Semrock) and collected by GaAsP or Multialkali photomultiplier tubes. Scanning was performed in a bidirectional mode with a resonant scanner at 400 Hz and 12 bit, with a zoom of 1x, 512 $\times$ 512 pixels. For optimal detection of blood vessels (CD31-AF647), a single tile scan was made at the last timepoint with InSight laser tuned at 1250 nm. At the postprocessing step, this image was co-registered to the time-lapse movie.

Breast tumor IVM imaging was performed as described before¹⁵ and data was kindly provided by Dr. Nienke Vriskoop. GBM IVM imaging was performed as described before⁷ and data was kindly provided by Prof. Jacco van Rheenen. Breast epithelial IVM imaging and cell tracking with MaSCOT-AI was performed as described before³¹ and single cell tracks were downloaded from Zenodo repository (14503476).

Co-registration of IVM imaged blood vessel imaging

CD31-AF647 tile scan featuring optimal visualization of blood vessels during live imaging was co-registered to the last timepoint of the time-lapse movie. Both images (last timepoint of time-lapse tile-scan and blood vessel tile-scan) were visually inspected to determine and annotate the internal landmarks using Imaris (Oxford Instruments), versions 9.5 to 9.6. These landmarks served as a reference for the co-registration software to overlap both images. To perform the co-registration, a modification of *elastix*⁶⁹ was used; a python-based open-source software based on Insight Segmentation and Registration Toolkit (ITK) plus SimpleElastix⁷⁰; a collection of different algorithms used on medical image registration. Due to the extensive number of potential parameter combinations, specific parameters were manually chosen. This selection was conducted using the Slicer3D visualization software, version 5.2.2.

Immunohistochemistry

3D immunohistochemistry was performed as described before⁴³. After the live imaging session, mouse brains were immersed in 5 ml 4% paraformaldehyde (PFA) pH 7.4 overnight on ice. After fixation, brains were blocked in 5-10 ml Wash Buffer 1 (2 ml Tween-20, 2 ml Triton X-100, 2 ml 10% SDS, and 2 g BSA in 1 l PBS) for 5 hours. For multiplex immunolabeling, a two- or three-round staining protocol was used as previously described^{42,43}. Washing and incubation steps were performed in Wash Buffer 2 (1 ml Triton X-100, 2 ml 10% SDS, and 2 g BSA in 1 l PBS). A thick (1 mm) slice of the cortex part that was under the cranial imaging window was dissected with a scalpel before proceeding to immunolabeling. The following primary antibodies were used: rabbit anti-Olig2 1:100 (AB9610, Merk), goat anti-Iba1 1:200 (NB100-1028SS, NovusBio) and CD31-AF647 1:250 (A14716, ThermoFisher). As secondary antibodies, donkey anti-rabbit AF405 1:250 (ab175651, Abcam) and donkey anti-goat AF633 1:250 (A21082, ThermoFisher) were used. After the last washing step, tissues were optically cleared by three stepwise incubations of an increasing concentration of FUnGI⁷¹ clearing agent (25%/50%/75%, diluted in PBS) for 1h at room temperature⁴². The final incubation step with 100% FUnGI was performed overnight at 4°C.

mLSR3D imaging

mLSR3D imaging^{42,43} of thick slices of fixed mouse cortex was performed using a Zeiss LSM880 system equipped with a 32-channel spectral detector. The signal from all fluorophores present in the samples was collected in a single acquisition and linear unmixing was used to obtain separate channels. Unmixing was carried out by the Online Fingerprinting mode of the microscope. Pre-acquired references of each single fluorophore were used to enable multispectral on-the-fly unmixing. Imaging was performed using a 25x multi-immersion objective 0.8 NA (Zeiss 420852-9871) with a working distance of 570µm. Tile scans were acquired using a voxel size of 0.332×0.332×1.200 µm, a pixel dwell of 2µs, and a 10% tile overlap.

Imaris image processing

DMG IVM datasets were processed using Imaris software (Oxford Instruments, versions 9.5 to 10.1) for 3D visualization, shift correction, object rendering, and cell tracking of time-lapse movies.

Imaris Cell tracking

The *Channel Arithmetics* Xtension was used for channel unmixing of overlapping signals. The Spots ImarisTrack module was used for object detection and semi-automated tracking of tumor cells (autoregressive motion). Tumor cell tracks were manually corrected to ensure accurate tracking. Around 50-100 cells per position were manually corrected and labeled as such for posterior selection of accurate tracks. To determine the direction of tumor cell movement relative to the tumor core, at the image edge closer to the tumor core a “Tumor edge” surface object was manually rendered using the contour tool. The *Distance Transformation* Xtension was used to measure the distance between tumor cells and the “Tumor edge”. For tracked tumor cells, time-lapse data containing the coordinates of each cell, the values of cell speed, mean square displacement, displacement delta length, displacement length, and distance to “Tumor edge” were exported. See <https://qbi.uq.edu.au/files/40820/ImarisManual.pdf> for statistical details.

Imaris Microenvironmental factors rendering

For rendering the microenvironmental factors detected during IVM or by post-IVM LSR-3D imaging, the Surface and Spots modules of Imaris were used. Surfaces were used for the SR101, CDr20, Olig2, DAPI, and Iba1 stainings, and Spots were used for CD31+ protruding or elongated structures, as previously described⁴⁴. For rendered structures, positional data containing the coordinates of objects were exported for large-scale spatial phenotyping with Cytomap.

Fiji image processing

μSAM cell segmentation

Segmentation was performed using the μSAM plugin for Napari³³, where *vit-b* model was selected to compute embeddings for each timeframe. Automatic segmentation was applied, and parameters were adjusted for optimal results. Manual corrections were made to remove unwanted surfaces or add missing cells. Once segmentation was complete, frames were combined into a single stack in ImageJ and tracked with Trackmate.

Fiji tracking

Breast tumor IVM data was manually tracked with Fiji MtrackJ plugin as previously described^{10,15}.

Adult glioma cells, segmented with μSAM, were consequently tracked with Fiji TrackMate plugin. The labeled image stack, segmented using μSAM, was imported into ImageJ for cell tracking with the Fiji TrackMate plugin. The “Label Image Detector” was employed to identify individual cells, and the LAP Tracker was used to link cells across timeframes. Tracking parameters were optimized to account for missing frames and ensure accurate cell trajectories. The statistics for each tracked cell were subsequently exported. To integrate both dynamic and morphological data, a custom script was developed to merge the dynamic information from TrackMate with the morphological data obtained from the μSAM segmentation labels for each individual cell. This integration was achieved by assigning the “TRACK_ID” from the TrackMate “allspots” output file to each segmented instance extracted from the TIFF file. The assignment was performed using the “measure” function in scikit-image and the KDTree distance function from the scipy module, based on the x, y, and z positions of the cells.

The script is available both as a downloadable version on GitHub (<https://github.com/imAlgene-Dream3D/MorphoTrack-merger>) and as an online application (<https://morphotrack-merger.streamlit.app/>), making this tool more accessible for non-coding users.

Tumor large-scale spatial phenotyping with Cytomap

The positional data of various environmental surface features, segmented from LSR-3D imaging data, was analyzed for large-scale spatial phenotyping using the Cytomap Spatial Analysis Toolbox⁴⁴. Briefly, neighborhoods with a 100 μm radius were generated. Neighborhoods were then clustered into 2 regions based on DMG cell positioning using the David Balwin algorithm. Using the gating tool, the tumor regions (containing DMG objects) were defined. Next, the neighborhoods were clustered again into 3 different regions based on the distribution of Iba1, Olig2, and CD31: *Void*, *TAMM/Oligo*, and *TAMM/vascularized*. By using the *Manually Defined Regions* option, only the tumor region was classified. Finally, for each region, fold-change values were calculated and exported using the *Region Statistics* tab. To map the assigned regions onto IVM movies, a 3D image of the cluster distribution within the tumor was generated and exported for each sample (Figure Supplement 5a). Next, regions within the IVM movies were visually matched to the corresponding regions identified by the Large-Scale Phenotyping module of Cytomap (Figure 3c). For each mouse, at least one or two representative positions per matched region type were selected, cropped, and analyzed to assess tumor cell behavior, following the previously described cell tracking methodology (*Imaris Cell tracking*).

BEHAV3D Tumor Profiler framework

BEHAV3D-TP was developed using the R Studio version 4.3.3. It features three modules: 1) *Heterogeneity module* with optional 2) *Large-scale and/or* 3) *Small-scale phenotyping module*. These modules are described in detail below, exemplified by the analysis of DMG data from this study. All of the modules are included in the same script available in Github [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler], but can be run independently from one another. To acquire this independency, the optional modules are directly combined to the necessary sections of the *Heterogeneity module*. Consequently, the user can use whichever module is best suited for their purposes. The complete workflow—combining heterogeneity analysis with large- and small-scale phenotyping—is primarily optimized for datasets processed with Imaris due to its wide use in the IVM community and special suitability for 3D visualization and tracking. In order to further facilitate public access to the BEHAV3D-TP, a Google Colab notebook has also been created accessible through the same Github page [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler]. This environment is friendly to anyone regardless of their coding expertise as it has a step-by-step follow-through execution through the pipeline.

In addition, to broaden accessibility and support alternative data processing pipelines, the core *Heterogeneity module* of BEHAV3D-TP is also compatible with Fiji-based tracking data, specifically outputs generated by TrackMate, MTrackJ and ManualTracking. A separate Google Colab notebook is provided for these workflows on the Github page, allowing users to easily import Fiji-processed datasets, adjust analysis parameters, identify different morpho-dynamic single cell patterns.

A Wiki demo run and a video tutorial (Supplementary Video 1) are also provided through the Github page [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler] for further clarification.

To ensure that your input data aligns with the workflow, we provide an online application for data quality control (https://github.com/alievkrash/BEHAV3D_TP_dataQC). This tool allows users to visualize various features, grouped by a condition of interest, and check for issues such as a high number of missing values (NA). It enables users to assess their data for any inconsistencies or unexpected patterns before proceeding with further analysis.

Heterogeneity module

To account for missing time points and to create a time series with the same time interval for each tumor cell time series, linear interpolation was used to estimate the values of missing time points. To compare time series independently, tracks were cropped to the minimal common length among all tracks. In the case of DMG data to a 2.6 hour duration. For tracked tumor cells, time-lapse data containing the coordinates of each cell, the values of cell speed, mean square displacement, displacement delta length, displacement length, and relative distance to “Tumor edge” were used to calculate a cross-distance matrix between time series. First, we performed a principal component (PC) analysis and selected the number of PCs that explained at least 90% of data variance (3).

The previously calculated PCs were then used to compute the cross-distance matrix between each tumor cell multivariate time series using the dynamic time warping algorithm from the package “dtwclust”. To visualize heterogenous tumor cell behaviors in 2 dimensions, dimensionality reduction on the cross-distance matrix was performed by the Uniform Manifold Approximation and Projection method (“umap” package). Clustering was performed using the k-means clustering algorithm. For each cluster, the average values of distinct behavioral and environmental components were calculated and plotted as a heatmap (“pheatmap”).

Large-scale phenotyping module

The information obtained from mLSR3D and Cytomap about large-scale TME regions were combined with the DMG cells behavioral information (see above section *Tumor large-scale spatial phenotyping with Cytomap*) to examine the large-scale features of tumor cells. For tumor cells, the mean speed, square displacement and raw movement were analyzed.

Small-scale phenotyping module

The positional coordinates of tracked tumor cells and detected microenvironmental factors were used to determine the small-scale landscape of tumor cells. For tumor cells, the average distance to the closest 10 neighbors was measured to determine tumor cell density. For each tumor cell and each environmental component, the number of microenvironmental objects in a 30- μ m radius and the distance to the closest object were computed. For each behavioral cluster, the values of closest neighbors, minimal distance to each environmental factor, and number of environmental objects in the 30 μ m radius were plotted and compared.

Statistical analyses

Statistical analyses were performed using R. The representation of n may refer to individual mice, distinct imaging positions within different mice, or individual cell tracks, as indicated in the figure legends. Two-tailed unpaired t-tests were performed for comparison between two groups unless indicated otherwise (Figure supplement 1d). For comparisons of more than two groups we performed a one-way ANOVA (Figure 2d [↗](#) and Figure 4b [↗](#)) followed by Tukey HSD correction (Figure Supplement 4c (p values in legend), Figure 3d [↗](#), Figure 4 c,e [↗](#), Figure Supplement b, e-g, Figure Supplement 6 c,d). To compare the distribution of different morphodynamic clusters of TEB cells, we pooled all positions and mice due to the low number of cells per position. We then constructed a contingency table and performed a Chi-square test (Figure supplement 2d). To compare the distance to the nearest neighbors between morphodynamic clusters of GBM cells, a mixed-effects linear regression was performed with mouse as a random effect (Figure supplement 3d).

Data availability

DMG imaging data is publicly available at BioStudies accession number S-BIAD1759 (<https://www.ebi.ac.uk/biostudies/studies/S-BIAD1759>). Breast cancer IVM data was previously described¹⁵ and kindly provided by Dr Nienke Vrisekoop. Adult glioma IVM data was previously described⁷ and kindly provided by Prof. Jacco van Rhennen. Healthy breast epithelium IVM was previously described³¹ and kindly provided by Prof. Jane Visvader. Demo outputs and Jupyter notebooks of Imaris- and Fiji-processed data are provided in our GitHub repository (https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler).

Code availability

We provide the BEHAV3D-TP framework in a Google Colab user-friendly environment [<https://colab.research.google.com/drive/1Jl7ysqFf3tvd6Df4YUsSZ8RbuXw8wba?usp=sharing>], as well as in an R script format in Github [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler]. Additionally, a Wiki [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler/wiki] to follow through a demo dataset as well as a video tutorial (Supplementary Video 1) has been developed for further assistance. For the BEHAV3D-TP heterogeneity module, we offer an online app to assess data quality (<https://behav3d-tp-data-qc.streamlit.app/>), with the corresponding code available on GitHub (https://github.com/alievakrash/BEHAV3D_TP_dataQC). Additionally, for quantifying morphological features derived from Microsam segmentation and integrating them with dynamic features from TrackMate, we provide another online app (<https://morphotrack-merger.streamlit.app/>) and related code on GitHub (<https://github.com/imAIgene-Dream3D/MorphoTrack-merger>).

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Additional files

Figure supplement 1. Example application of BEHAV3D-TP for 2D movement pattern analysis of breast cancer cells tracked with MTrackJ Fiji plugin. **a)** Representative intravitaly imaged position showing breast tumor cells tracked in 2D with the MTrackJ plugin. Red: MDA-MB-231 cells; Blue: Second Harmonic Generation; Multicolored tracks: individually tracked cells. **b)** UMAP plot showing six color-coded breast tumor cell behavioral clusters identified by BEHAV3D-TP. Each datapoint represents one cell track. A total of 935 individual cells were tracked. **c)** Heatmap depicting relative values of breast tumor cell dynamic and spatial features indicated for each cluster, named according to their most distinct characteristics.

Arbitrary units in respect to maximal and minimal values for each feature. See Table 1 for features description. **d)** Boxplots showing the percentages of different breast cancer behavioral clusters for 4T1 and MDA-MB-231 cells. Each point represents a different imaging position. P-values represent the significance between both conditions measured with a Student's T test. **e)** Breast tumor cell tracks classified according to behavior (color coded by cluster in c). In all panels: representative of $n = 13$ independent positions from $n=4$ mice. [🔗](#)

Figure supplement 2. Example application of BEHAV3D-TP for 2D morphodynamic analysis of cells segmented with CellPose2.0 and tracked using TrackMate. **a)** Schematic overview of the segmentation and tracking pipeline using CellPose2.0 and TrackMate (see Methods). **b)** UMAP plot showing four color-coded morphodynamic clusters of healthy breast epithelial cells from TEBs identified by BEHAV3D-TP. Each datapoint represents one cell track. A total of 291 individual cells were tracked. **c)** Heatmap depicting relative values of epithelial cell dynamic and morphological features indicated for each cluster, named according to their most distinct characteristics. Arbitrary units in respect to maximal and minimal values for each feature. **d)** Stacked bar plot showing the distribution of morphodynamic cell types across the two indicated TEB populations: HR+ luminal progenitor cells and HR- luminal progenitor cells. Combined data from 19 mice and 72 independent positions. Chi-square test indicates $p=0.02$ for cluster 2. [🔗](#)

Figure Supplement 3. Figure supplement 3: Example application of BEHAV3D-TP for 3D morphodynamic analysis of cells segmented with μ SAM and tracked using TrackMate. **a)** Schematic overview of the segmentation and tracking pipeline using μ SAM and TrackMate. An online app was developed to integrate morphological and dynamic features (see Methods). **b)** UMAP plot showing four color-coded morphodynamic clusters of GBM cells identified by BEHAV3D-TP. Each datapoint represents one cell track. Total of 460 individually tracked cells. **c)** Heatmap depicting relative values of GBM cell dynamic and morphological features indicated for each cluster, named according to their most distinct characteristics. Arbitrary units in respect to maximal and minimal values for each feature. See Table 1 for features description. **d)** Boxplot showing Average distance to the 3 closest GBM neighbors (pixels) across distinct GBM morphodynamic clusters. Each point represents an individual tracked cell, with color indicating the imaged position ($n = 6$) and shape representing the mouse ($n = 4$). P-values denote statistical significance across behavioral clusters, assessed using a linear mixed model with mouse as a random effect, followed by Tukey's post hoc test. $P < 0.001$ (***). [🔗](#)

Figure supplement 4. DMG cells heterogeneity captured by intravital imaging. **a,** Schematic representation of an intravital imaging experimental design, capturing timelapses up to 5.4 hours of DMG cells expressing H2BmNeonGreen and mScarlet proteins, followed by single-cell tracking. **b,** Representative intravital imaged position showing DMG cell tracks colored by behavioral cluster projected over the last timepoint of the original IVM time-lapse. **c,** Quantification of average DMG cell speed ($\mu\text{m}/\text{h}$) in each identified behavioral cluster. Significance of differences across behavioral clusters tested through ANOVA followed by Tukey's post hoc test. All comparisons significant, except cluster 3-4 and 6-4. **d,** Behavioral cluster frequency distribution across individual IVM imaged mice. Each point indicates an individual cell track, color depicts $n = 18$ independent positions; shape depicts $n=6$ mice. [🔗](#)

Figure supplement 5. Cell type and behavioral cluster frequency distribution in identified TME large-scale regions. **a,** Representative CytoMap output showcasing, 3D spatial distribution of DMG cells in distinct TME large-scale regions: Void (blue), TAMM/Oligo (yellow) and TAMM/vascularized (red). Each datapoint represent a unique DMG cell within a classified large-scale region. **b, c,** Fold change of cell count of DMG, Iba1+, Olig2+ and CD31+ cell types, identified with mLSR3D fixed imaging, relative to the mean cell count in each TME large scale region (Void, TAMM/Oligo, TAMM/vascularized), both in boxplot (**b**) and table (**c**) formats. In (**b**) each point represents a mouse, $n=7$. For DMG cell count: TAMM/Oligo versus Void, $P < 0.01$ (**); TAMM/Oligo versus TAMM/vascularized, $P < 0.05$ (*). For Iba1 cell count: TAMM/Oligo versus Void, $P < 0.001$ (***); TAMM/vascularized versus Void, $P < 0.001$ (***). In Olig2 cell count: TAMM/Oligo versus Void, $P < 0.001$ (***); TAMM/Oligo versus TAMM/vascularized, $P < 0.001$ (***). For CD31 cell count: TAMM/Oligo versus Void, $P < 0.001$ (***); TAMM/vascularized versus Void, $P < 0.001$ (***); TAMM/Oligo versus TAMM/vascularized, $P < 0.001$ (***). **d,** Boxplots showing the percentages of different DMG behavioral

clusters in Void, TAMM/Oligo and TAMM/vascularized TME large-scale regions. Each point represents a different imaging position. Color: $n = 18$ independent positions; Shape: $n=6$ mice. **e, f, g**, Violin plots showing DMG single cell features (Tumor cell speed ($\mu\text{m}/\text{hour}$) (**e**), Average speed displacement (**f**) and Tumor cell velocity ($\mu\text{m}/\text{hour}$) (**g**)) differences among Void, TAMM/Oligo and TAMM/vascularized TME large-scale regions. **e**, TAMM/vascularized versus Void, $P < 0.01$ (**); TAMM/vascularized versus TAMM/Oligo, $P < 0.01$ (**). **g**, Tumor cell velocity can be either towards brain core (down, negative values) or towards brain parenchyma (up, positive values). For **e, f, g**: $n = 18$ independent positions from $n=6$ mice. [🔗](#)

Figure supplement 6. Cell distribution and behavioral clusters in small-scale TME. **a, b**, Representative 3D IVM rendering of DMG nuclei (cyan), CD31+ cells (grey) and SR101+ (**a**, red) or CD20r+ cells (**b**, red). Scale bars, $100\ \mu\text{m}$. **c, d**, Boxplot showing Average distance to the 10 closest DMG neighbors (μm) (**c**) and Number of SR101+ cells in a $30\ \mu\text{m}$ radius (**d**) across distinct DMG single-cell behavioral clusters. Each point represents an individual tracked cell, color depicts the imaged position and the shape denotes the mouse. P-values represent the significance of differences across behavioral clusters tested through ANOVA followed by Tukey's post hoc test. $P < 0.05$ (*); $P < 0.01$ (**); $P < 0.1$ (-). **c**, $n=18$ independent positions imaged in 6 mice. **d**, $n=8$ independent positions imaged in 3 mice. [🔗](#)

Table 1. [🔗](#)

Supplementary video 1. BEHAV3D Tumor Profiler pipeline main characteristics and walkthrough. Summary video of BEHAV3D TP pipeline. [🔗](#)

Supplementary video 2. Representative intravital time-lapse video of DMG cells. Raw DMG cells time-lapse (right panel). Cell tracks were classified according to DMG behavior (right panel, color coded by cluster). Representative movie from $n=18$ independent positions imaged in 6 mice. Scale bars $40\ \mu\text{m}$. [🔗](#)

Supplementary video 3. Dynamic multispectral representative time-lapse video of DMG cells and CD20r+ TAMM. Time-lapse showing the movement of DMG cells (grey) relative to CD20r+ TAMM (blue). The trajectories towards the TAMM of two DMG cells show a path from the initial (blue) to the final position (red). Left: Tumor edge; right: tumor core. Representative movie from $n=10$ independent positions imaged in 3 mice. Scale bars $10\ \mu\text{m}$, $7\ \mu\text{m}$ after zoom in. [🔗](#)

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Reviewer #1 (Public review):

Summary:

Intravital microscopy (IVM) is a powerful tool that facilitates live imaging of individual cells over time in vivo in their native 3D tissue environment. Extracting and analysing multi-parametric data from IVM images however is challenging, particularly for researchers with limited programming and image analysis skills. In this work, Rios-Jimenez and Zomer et al have developed a 'zero-code' accessible computational framework (BEHAV3D-Tumour Profiler) designed to facilitate unbiased analysis of IVM data to investigate tumour cell dynamics (via the tool's central 'heterogeneity module') and their interactions with the tumour microenvironment (via the 'large-scale phenotyping' and 'small-scale phenotyping' modules). It is designed as an open-source modular Jupyter Notebook with a user-friendly graphical user interface and can be implemented with Google Colab, facilitating efficient, cloud-based computational analysis at no cost. Demo datasets are also available on the authors GitHub repository to aid user training and enhance the usability of the developed pipeline.

To demonstrate the utility of BEHAV3D-TP, they apply the pipeline to timelapse IVM imaging datasets to investigate the in vivo migratory behaviour of fluorescently labelled DMG cells in tumour bearing mice. Using the tool's 'heterogeneity module' they were able to identify distinct single-cell behavioural patterns (based on multiple parameters such as directionality, speed, displacement, distance from tumour edge) which was used to group cells into distinct categories (e.g. retreating, invasive, static, erratic). They next applied the framework's 'large-scale phenotyping' and 'small-scale phenotyping' modules to investigate whether the tumour microenvironment (TME) may influence the distinct migratory behaviours identified. To achieve this, they combine TME visualisation in vivo during IVM (using fluorescent probes to label distinct TME components) or ex vivo after IVM (by large-scale imaging of harvested, immunostained tumours) to correlate different tumour behavioural patterns with the composition of the TME. They conclude that this tool has helped reveal links between TME composition (e.g. degree of vascularisation, presence of tumour-associated macrophages) and the invasiveness and directionality of tumour cells, which would have been challenging to identify when analysing single kinetic parameters in isolation.

The authors also evaluated the BEHAV3D TP heterogeneity module using available IVM datasets of distinct breast cancer cell lines transplanted in vivo, as well as healthy mammary epithelial cells to test its usability in non-tumour contexts where the migratory phenotypes of cells may be more subtle. This generated data is consistent with that produced during the original studies, as well as providing some additional (albeit preliminary) insights above that previously reported. Collectively, this provides some confidence in BEHAV3D TP's ability to uncover complex, multi-parametric cellular behaviours that may be missed using traditional approaches.

Overall, this computational framework appears to represent a useful and comparatively user-friendly tool to analyse dynamic multi-parametric data to help identify patterns in cell migratory behaviours, and to assess whether these behaviours might be influenced by neighbouring cells and structures in their microenvironment. When combined with other methods, it therefore has the potential to be a valuable addition to a researcher's IVM analysis 'tool-box'.

Strengths:

- Figures are clearly presented, and the manuscript is easy to follow.
- The pipeline appears to be intuitive and user-friendly for researchers with limited computational expertise. A detailed step-by-step video and demo datasets are also included to support its uptake.
- The different computational modules have been tested using relevant datasets, including imaging data of normal and tumour cells in vivo.
- All code is open source, and the pipeline can be implemented with Google Colab.
- The tool combines multiple dynamic parameters extracted from timelapse IVM images to identify single-cell behavioural patterns and to cluster cells into distinct groups sharing similar behaviours, and provides avenues to map these onto in vivo or ex vivo imaging data of the tumour microenvironment

Weaknesses:

- The tool does not facilitate the extraction of quantitative kinetic cellular parameters (e.g. speed, directionality, persistence and displacement) from intravital images. To use the tool researchers must first extract dynamic cellular parameters from their IVM datasets using other software including Imaris, which is expensive and therefore not available to all. Nonetheless, the authors have developed their tool to facilitate the integration of other data formats generated by open-source Fiji plugins (e.g. TrackMate, MTrackJ, ManualTracking) which will help ensure its accessibility to a broader range of researchers.
- The analysis provides only preliminary evidence in support of the authors conclusions on DMG cell migratory behaviours and their relationship with components of the tumour microenvironment. The authors acknowledge this however, and conclusions are appropriately tempered in the absence of additional experiments and controls.

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Reviewer #2 (Public review):

Summary:

The authors produce a new tool, BEHAV3D to analyse tracking data and to integrate these analyses with large and small scale architectural features of the tissue. This is similar to several other published methods to analyse spatio-temporal data, however, the connection to tissue features is a nice addition, as is the lack of requirement for coding. The tool is then used to analyse tracking data of tumour cells in diffuse midline glioma. They suggest 7 clusters exist within these tracks and that they differ spatially. They ultimately suggest that there these behaviours occur in distinct spatial areas as determined by CytoMAP.

Strengths:

- The tool appears relatively user-friendly and is open source. The combination with CytoMAP represents a nice option for researchers.

- The identification of associations between cell track phenotype and spatial features is exciting and the diffuse midline glioma data nicely demonstrates how this could be used.

Weaknesses:

- The revision has dealt with many concerns, however, the statistics generated by the process are still flawed. While the statistics have been clarified within the legends and this is a great improvement in terms of clarity the underlying assumptions of the tests used are violated. The problem is that individual imaging positions or tracks are treated as independent and then analysed by ANOVA. As separate imaging positions within the same mouse are not independent, nor are individual cells within a single mouse, this makes the statistical analyses inappropriate. For a deeper analysis of this that is feasible within a review please see Lord, Samuel J., et al. "SuperPlots: Communicating reproducibility and variability in cell biology." *The Journal of cell biology* 219.6 (2020): e202001064. Ultimately, while this is a neat piece of software facilitating the analysis of complex data, the fact that it will produce flawed statistical analysis is a major problem. This problem is compounded by the fact that much imaging analysis has been analysed in this inappropriate manner in the past, leading to issues of interpretation and ultimately reproducibility.

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Reviewer #3 (Public review):

The manuscript by Rios-Jimenez developed a software tool, BEHAV3D Tumor Profiler, to analyze 3D intravital imaging data and identify distinctive tumor cell migratory phenotypes based on the quantified 3D image data. Moreover, the heterogeneity module in this software tool can correlate the different cell migration phenotypes with variable features of the tumor microenvironment. Overall, this is a useful tool for intravital imaging data analysis and its open-source nature makes it accessible to all interested users.

Strengths:

An open-source software tool that can quantify cell migratory dynamics from intravital imaging data and identify distinctive migratory phenotypes that correlate with variable features of the tumor microenvironment.

Weaknesses:

Motility is only one tumor cell feature and is probably not sufficient to characterize and identify the heterogeneity of the tumor cell population that impacts their behaviors in the complex tumor microenvironment (TME). For instance, there are important non-tumor cell types in the TME, and the interaction dynamics of tumor cells with other cell types, e.g., fibroblasts and distinct immune cells, play a crucial role in regulating tumor behaviors. BEHAV3D-TP focuses on only motility feature analysis, and cannot be applied to analyze other tumor cell dynamic features or cell-cell interaction dynamics.

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Author response:

The following is the authors' response to the original reviews

We thank the reviewers for their positive and constructive comments on the manuscript. In the revised manuscript we addressed these comments, which we believe have improved the

quality of our work.

In summary:

(1) We acknowledge the reviewer's suggestion to incorporate open-source segmentation and tracking functionalities, increasing its accessibility to a wider user base; however, these additions fall outside the primary scope of our current work, which is to provide an analytical framework for IVM data after segmentation and tracking. Developing open-source segmentation and tracking tools represents a substantial undertaking in its own right, which has been comprehensively explored in other studies (e.g. <https://doi.org/10.4049/jimmunol.2100811>; <https://doi.org/10.7554/eLife.60547>; <https://doi.org/10.1016/j.media.2022.102358>; <https://doi.org/10.1038/s41592024-02295-6> - now cited in our revised manuscript).

In our analyses, we used data processed with Imaris, a commercial software that, despite its limitations, is widely used by the intravital microscopy community due to its user-friendly platform for 3D image visualization and analysis. Nevertheless, recognizing the need for compatibility with tracking data from various pipelines, we have modified our tool to accept other data formats, such as those generated by open-source Fiji plugins like TrackMate, MTrackJ, ManualTracking (https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#data-input). These updates are available in our GitHub repository and are described in the revised manuscript.

(2) We appreciate the reviewer #3 suggestion to incorporate additional features into our analytical pipeline. In response, we have already updated the GitHub repository to allow users to input and select which features (dynamic, morphological, or spatial) they wish to include in the analysis (https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#feature-selection). In the revised manuscript, we highlighted this new functionality and provided examples using alternative datasets to demonstrate the application of these features.

(3) We appreciate the constructive feedback of reviewers #1 and #2 regarding the statistical analysis and interpretation of the data presented in Figures 3 and 4. We understand the importance of clarity and rigor in data analysis and presentation, and we addressed the concerns raised in the revised version of the manuscript.

(4) We appreciate reviewer #1's suggestion regarding the inclusion of demo data, as we believe it would greatly enhance the usability of our pipeline. We acknowledge that this was an oversight on our part. To address this, we have now added demos to our GitHub repository (https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler/tree/BEHAV3D_TP-v2.0/demo_datasets).

In the revised manuscript, we referenced this addition and present new figures with examples of these demo's processing different IVM dataset (2D/3D, different tumors and healthy tissues). Additionally, we have provided processed DMG IVM movie samples in an imaging repository.

(5) Finally, we made some small changes to the manuscript based on the reviewers' feedback.

Below we provide a point-by-point response to the reviewers' comments

Reviewer #1 (Public review):

Comment #1: A key limitation of the pipeline is that it does not overcome the main challenges and bottlenecks associated with processing and extracting quantitative cellular data from timelapse and longitudinal intravital images. This includes correcting breathing-induced movement artifacts, automated registration of longitudinal images taken over days/weeks, and accurate, automated segmentation and tracking of individual cells over time. Indeed, there are currently no standardised computational

methods available for IVM data processing and analysis, with most laboratories relying on custom-built solutions or manual methods. This isn't made explicit in the manuscript early on (described below), and the researchers rely on expensive software packages such as IMARIS for image processing and data extraction to feed the required parameters into their pipeline. This limitation unfortunately reduces the likely impact of BEHAV3D-TP on the IVM field.

As highlighted above, the tool does not facilitate the extraction of quantitative kinetic cellular parameters (e.g. speed, directionality, persistence, and displacement) from intravital images. Indeed, to use the tool researchers must first extract dynamic cellular parameters from their IVM datasets, requiring access to expensive software (e.g. IMARIS as used here) and/or above-average computational expertise to develop and use custom-made open-source solutions. This limitation is not made explicit or discussed in the text.

We acknowledge the reviewer's suggestion to incorporate open-source segmentation and tracking functionalities, increasing its accessibility to a wider user base; however, these additions fall outside the primary scope of our current work and represent a substantial undertaking in their own right. Several studies (e.g., Diego Ulisse Pizzagalli et al., J Immunol (2022); Aby Joseph et al., eLife (2020); Molina-Moreno et al., Medical Image Analysis (2022); Hidalgo-Cenalmor et al., Nat Methods (2024); Ershov et al., Nat Methods (2022)) have comprehensively addressed these topics, and we now reference them in the revised manuscript to provide readers with relevant background.

The objective of our manuscript is not to develop a complete segmentation or tracking pipeline but rather to introduce an analytical framework capable of extracting enhanced insights from the data generated by existing tools. This goal arises from our observations of the field: despite significant investment in image processing, researchers often rely on simplistic approaches, such as averaging single parameters across conditions, which can obscure tumor heterogeneity and spatial behavioral dynamics within the tumor microenvironment.

Our current tool focuses on providing this much-needed analytical capability. For our analysis we used Imaris, a widely utilized software in the intravital microscopy (IVM) community, known for its intuitive 3D visualization and analysis platform despite certain limitations.

In our own literature search of recent IVM studies published by leading laboratories in high-impact journals, we found that close to half used Imaris, while the remainder primarily relied on manual workflows with Fiji plugins. Thus, we consider it valuable to offer a pipeline compatible with such commonly used software, given its prevalence in the field.

However, following the suggestion of the reviewer, and to enhance the tool's flexibility and compatibility, we have expanded the pipeline to accept data formats generated by open-source Fiji plugins, such as TrackMate, MTrackJ, and ManualTracking. These updates are detailed in the revised manuscript and are implemented in our GitHub repository (https://github.com/imAIGene-Dream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#data-input), where we also provide several demos using TrackMate and Imaris processed data. This addition demonstrates our tool's capability to integrate with segmented and tracked datasets from diverse platforms, increasing its applicability to a broader range of researchers using both commercial and open-source pipelines.

Comment #2: The number of cells (e.g. per behavioural cluster), and the number of independent mice, represented in each result figure, is not included in the figure legends and are difficult to ascertain from the methods.

We appreciate the reviewer's constructive feedback regarding the clarity of the number and type of replicates used in our analyses. In the revised manuscript, we have included detailed information in the figure legends and the number of independent mice represented in each figure legend to ensure transparency. Regarding the number

of cells, we have indicated the total number of processed cells in Figure 2b legend (953 cells). Additionally, we have now included figures (Sup Fig 4c, Sup Fig 5e-g, Fig 5c,e, Sup Fig 6 c,d) for each cluster, where individual dots represent the individual cell tracks with color indicating the position and the shape indicating individual mice.

Comment #3: The data used to test the pipeline in this manuscript is currently not available, making it difficult to assess its usability. It would be important to include this for researchers to use as a 'training dataset'.

As stated above we acknowledge that this was an oversight on our part and thank the reviewer for pointing this out. To address this, we have now added demo data to our GitHub repository (BEHAV3D_Tumor_Profiler/demo_datasets at main · imAIgeneDream3D/BEHAV3D_Tumor_Profiler · GitHub). In the revised manuscript we have referenced this addition in the Data availability section. Since we included now processing with Fiji as well, we provide 4 demo datasets (https://github.com/imAIgeneDream3D/BEHAV3D_Tumor_Profiler/tree/main/demo_datasets), one processed with Imaris in 3D; and one with CellPose2.0 and Trackmate in 2D; one processed with µSAM and Trackmate in 3D and one manually processed with MtrackJ in 2D. Moreover, we now provide Imaris-processed DMG IVM movie samples in an open-source repository.

Comment #4: Precisely how the BEHAV3D-TP large-scale phenotyping module can map large-scale spatial phenotyping data generated using LSR-3D imaging data and Cytomap to 3D intravital imaging movies is unclear. Further details in the text and methods would be beneficial to aid understanding.

We appreciate the reviewer's comment and in the revised manuscript we have now provided details in the methods section "Tumor large-scale spatial phenotyping with Cytomap" to clarify how the BEHAV3D-TP module maps LSR-3D and Cytomap data to 3D intravital imaging movies:

"To map the assigned regions onto IVM movies, a 3D image of the cluster distribution within the tumor was generated and exported for each sample (Figure Supplement 5a). Next, regions within the IVM movies were visually matched to the corresponding regions identified by the Large-Scale Phenotyping module of Cytomap (Figure 3c). For each mouse, at least one or two representative positions per matched region type were selected, cropped, and analyzed to assess tumor cell behavior, following the previously described cell tracking methodology (Imaris Cell tracking)."

Moreover, we updated Figure 3 c to further clarify these steps.

Comment #5: The analysis provides only preliminary evidence in support of the authors' conclusions on DMG cell migratory behaviours and their relationship with components of the tumour microenvironment. Conclusions should therefore be tempered in the absence of additional experiments and controls.

We appreciate the reviewer's comment and acknowledge that our conclusions should be tempered due to the preliminary nature of our evidence. In the revised version of the manuscript we have revised our conclusions accordingly and emphasize the necessity for

additional experiments and controls to further validate our findings on DMG cell migratory behaviors and their relationship with the tumor microenvironment.

In discussion: “While our findings suggest that microenvironmental factors may influence tumor cell migration, further studies will be necessary to establish causal relationships. Additional experimental validation, such as macrophage ablation experiments, could help clarify the specific contributions of these factors.”

Reviewer #1 (Recommendations for the authors):

(1) To test the ability of the pipeline to identify relevant patterns of migratory behaviours additional 'control' experiments would be helpful e.g. comparing non-invasive vs invasive tumour cell lines, artificially controlling migratory behaviours of cells such as implanting beads soaked in factors that would attract/repel cells?

(2) Does the pipeline work well for a variety of cell types/contexts? e.g. can it identify and cluster more subtle migratory behaviours such as non-tumour cells during tissue development or regeneration conditions?

We appreciate the reviewer's valuable suggestions. In the revised manuscript, we have included additional examples demonstrating the capability of our pipeline to investigate heterogeneous cell behavior across two additional experimental setups:

(1) We have now evaluated our BEHAV3D TP heterogeneity module using IVM data from breast cancer cell lines with varying migratory capacities (DOI: 10.1016/j.yexcr.2019.04.009). In these datasets, our pipeline extends beyond predefined characteristics based solely on speed, enabling the identification of distinct cell populations. Notably, our analysis reveals that the breast cancer lines exhibit different proportions of different migratory behaviors such as Fast, Intermediate, Very slow and Static (Supplementary Fig 1).

(2) We have now evaluated our BEHAV3D TP heterogeneity module using IVM data from healthy breast epithelial cells (DOI: 10.1016/j.celrep.2024.115073), where we identify distinct morphodynamic epithelial cell populations in the terminal end but of the mammary gland that have a distinct distribution among Hormone receptor (HR) + and HR- terminal end but cells.

(3) To support biological conclusions could the authors show that ablating tumour associated macrophages or vasculature alters the migratory patterns of nearby tumour cells?

We appreciate the reviewer's suggestion regarding the potential effects of ablating tumor-associated macrophages or vasculature on the migratory patterns of nearby tumor cells. While these experiments would functionally validate the observations made by our method, we would like to clarify that the primary focus of our study was on the development and application of computational tools for behavioral analysis and thus we consider that delving deeper in understanding the biology behind our observation is out of the scope of the current study. However, as mentioned previously, we have carefully tempered our conclusions to acknowledge the limitations of our current study. In the revised manuscript, we explicitly highlight that experiments involving the ablation of tumor-associated macrophages or vasculature would be crucial for further understanding the biological relevance of our findings.

Minor corrections to text:

(4) Line 63 - are references formatted correctly?

Thank you for pointing out this error. We have corrected it in the revised manuscript.

(5) Lines 161 -162 - 'intravitaly imaged' used twice in a sentence.

Thank you for pointing out the typo. We have corrected it in the revised manuscript.

Reviewer #2 (Public review):

Comment#1: The strength of democratizing this kind of analysis is undercut by the reliance upon Imaris for segmentation, so it would be nice if this was changed to an open-source option for track generation.

As noted in our previous response to Reviewer #1, we would like to point out that although Imaris is a commercial software, it is widely used in the intravital microscopy community due to its user-friendly interface. We conducted a literature review to evaluate this aspect and below we include references from leading laboratories in the IVM field that utilize Imaris. One of its key advantages, which we also utilized, is semi-automated data tracking that allows for manual corrections in 3D—a process that can be more challenging in other open-source software with less effective data visualization.

However, we recognize that enhancing our pipeline's compatibility with open-source options is important. To this end, we have updated our tool to support 2D and 3D data formats generated by open-source Fiji plugins like TrackMate, MTrackJ, and ManualTracking, improving compatibility with various segmentation and tracking pipelines (https://github.com/imAlgene-Dream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#data-input). In the revised manuscript, we describe the new functionality and demonstrate the operation of the BEHAV3D-TP heterogeneity module across various IVM datasets, processed in both 2D and 3D with different processing pipelines (Supplementary Fig 1-3). This includes CellPose 2.0 and the novel 'Segment Anything' model, followed by TrackMate tracking, applied to both tumor and healthy IVM data. Moreover we have developed a new web application that integrates morphological and tracking information from Segment Anything segmentation and Trackmate tracking, depicted in Supplementary Fig 3 a (<https://morphotracker-merger.streamlit.app/>). Additionally, we have updated the introduction to better clarify the scope of our study and include references to existing image processing solutions.

Comment#2: The main issue is with the interpretation of the biological data in Figure 3 where ANOVA was used to analyse the proportional distribution of different clusters. Firstly the n is not listed so it is unclear if this represents an n of 3 where each mouse is an individual or whether each track is being treated as a test unit. If the latter this is seriously flawed as these tracks can't be treated as independent. Also, a more appropriate test would be something like a Chi-squared test or Fisher's exact test. Also, no error bars are included on the stacked bar graphs making interpretation impossible. Ultimately this is severely flawed and also appears to show very small differences which may be statistically different but may not represent biologically important findings. This would need further study.

We appreciate the reviewer's insightful comments regarding the interpretation of the biological data in Figure 3.

To clarify, each imaged position is considered an independent biological replicate (n = 18 from a total of 6 mice). We acknowledge that the description of the statistical methods and the experimental units was not sufficiently clear in the previous version. In our original submission, we used an ANOVA to test whether the proportion of each behavioral cluster differed across the tumor microenvironment regions. Post hoc pairwise comparisons were

performed using Tukey's test, with the results shown in Supplementary Figure 2d (currently Fig 3d). However, we agree with the reviewer that this approach may be misleading when paired with stacked bar plots that lack error bars, as it can obscure individual variability and does not explicitly represent statistical uncertainty.

In the revised manuscript, we present the data as boxplots with individual data points, where each dot represents an imaged position, and the shape corresponds to a specific mouse. In Figure 3 d the y-axis displays the normalized percentage of each cluster across TME regions, expressed as z-scores. This normalization corrects for inter-mouse variability and facilitates a comparison of the relative distribution of clusters across TME regions, independent of the overall abundance differences between mice. We performed an ANOVA with Tukey's post hoc test for each individual behavioral cluster to assess differences across TME regions. Additionally, for transparency, in Supplementary Figure 5 d we provide the raw percentage values. The legends provide the number of positions and mice included in the analysis.

Comment#3: Figure 4 has similar statistical issues in that the n is not listed and, again, it is unclear whether they are treating each cell track as independent which, again, would be inappropriate. The best practice for this type of data would be the use of super plots as outlined in Lord et al. (2020) JCI - SuperPlots: Communicating reproducibility and variability in cell biology.

We appreciate the reviewer's comments and suggestions regarding Figure 4. In this case as we are comparing overall the behavioral clusters features, each individual cell is treated as a unit. In the revised manuscript, we have clarified this point in the figure legend and incorporated plots in Figure 4c and 4e, indicating the mouse and imaging position each data point originates from. This enhances the visualization of reproducibility and variability in our data, demonstrating that the results are consistent across multiple mice and positions and are not driven by a single mouse or imaging position.

Comment#4: The main issue that this raises is that the large-scale phenotyping module and the heterogeneity module appear designed to produce these statistical analyses that are used in these figures and, if they are based on the assumption that each track is independent, then this will produce inappropriate analyses as a default.

We appreciate the reviewer's comment, although we are unclear about the specific concern being raised. To clarify, in our large-scale phenotyping analysis, each position is assigned to a TME niche based on the CytoMAP analysis and the workflow outlined in Figure 3c. Multiple positions are imaged per mouse. For each position, we measure the proportion of tumor cells exhibiting a specific behavioral phenotype, and these proportions are subsequently used for statistical analysis (Figure 3 d).

In contrast, in Supplementary Fig. 5e-g, we treat each cell track as an individual unit, grouping them by their assigned large-scale region. Here, we assess whether differences between regions can be detected using a conventional single-feature analysis—a more traditional approach. However, we find that this method loses important behavioral patterns and distinctions that BEHAV3D-TP captures.

We hope that this explanation, along with the modifications made to the figures and figure legends, provides greater clarity.

Reviewer #3 (Public review):

Comment #1: The most challenging task of analyzing 3D time-lapse imaging data is to accurately segment and track the individual cells in 3D over a long time duration. BEHAV3D Tumor Profiler did not provide any new advancement in this regard, and

instead relies on commercial software, Imaris, for this critical step. Imaris is known to have a very high error rate when used for analyzing 3D time-lapse data. In the Methods section, the authors themselves stated that "Tumor cell tracks were manually corrected to ensure accurate tracking". Based on our own experience of using Imaris, such manual correction is tedious and often required for every time step of the movie. Therefore, Imaris is not a satisfactory tool for analyzing 3D time-lapse data. Moreover, Imaris is expensive and many research labs probably can't afford to buy it. The fact that BEHAV3D Tumor Profiler critically depends on the faulty ImarisTrack module makes it unclear whether the BEHAV3D tool or the results are reliable.

If the authors want to "democratize the analysis of heterogeneous cancer cell behaviors", they should perform image segmentation and tracking using open-source codes (e.g., Cellpose, Stardisk & 3DCellTracker) and not rely on the expensive and inaccurate ImarisTrack Module for the image analysis step of BEHAV3D.

We appreciate the reviewer's comments on the challenges of segmenting and tracking individual cells in 3D time-lapse imaging data. As mentioned previously (please refer to comment #1 to reviewer #1), our primary focus is to develop an analytical tool for comprehensive data analysis rather than developing tools for image processing. However to enhance accessibility, we have updated our tool to support data formats from open-source Fiji plugins, such as TrackMate, which will benefit users without access to commercial software (https://github.com/imAIgeneDream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#data-input). In Supplementary Figures 1, 2, and 3, we present IVM data from different sources, processed using three distinct methods: MTrackJ (Supplementary Fig. 1), Cellpose + TrackMate (Supplementary Fig. 2), and μ SAM + TrackMate (Supplementary Fig. 3). The latter two represent state-of-the-art deep learning approaches.

On the other hand, while we recognize the limitations of Imaris, it remains widely used in the intravital microscopy community due to its user-friendly interface for 3D visualization and semi-automated segmentation capabilities. Since no perfect tracking method currently exists, we initially utilized Imaris for its ability to allow manual correction of faulty tracks, ensuring the reliability of our results. This approach, not only widely used (see above) but was the best available option when we began our analysis, allowing us to obtain accurate results efficiently.

In the revised manuscript, we clarify the scope of our study and provide information on both Imaris and alternative processing options to strengthen the reliability of our findings:

In introduction: "While significant efforts have been made to develop opensource segmentation and tracking tools for live imaging data, including IVM22–27 fewer tools exist for the unbiased analysis of tumor dynamics. One major barrier is that implementing such analytical methods often requires substantial computational expertise, limiting accessibility for many biomedical researchers conducting IVM experiments. To bridge this gap, we present BEHAV3D Tumor Profiler (BEHAV3D-TP) by providing a robust, user-friendly tool that allows researchers to extract meaningful insights from dynamic cellular behaviors without requiring advanced programming skills."

In the Methods, we describe now describe not only Imaris processing pipeline, but also the μ SAM segmentation pipelines and reference to CellPose IVM processing, which are combined with TrackMate for tracking. Additionally, to integrate morphological information from μ SAM with tracking data from TrackMate, we developed a web tool to merge the outputs from both processing steps: <https://morphotrack-merger.streamlit.app/>

Comment #2: The authors developed a "Heterogeneity module" to extract distinctive tumor migratory phenotypes from the cell tracks quantified by Imaris. The cell tracks of the individual tumor cells are all quite short, indicating relatively low motility of the

tumor cells. It's unclear whether such short migratory tracks are sufficient to warrant the PCA analysis to identify the 7 distinctive migratory phenotypes shown in Figure 2d. It's also unclear whether these 7 migratory phenotypes correspond to unique functional phenotypes.

For the 7 distinctive motility clusters, the authors should provide a more detailed analysis of the differences between them. It's unclear whether the difference in retreating, slow retreating, erratic, static, slow, slow invading, and invading correspond to functional difference of the tumor cells.

While some tumor cells exhibit limited motility, indicated by short tracks, others demonstrate significant migratory capabilities (Figure 2 Invading and Retreating cells). This variability in tumor cell behavior is a central focus of our analysis, and our tool is specifically designed to identify and distinguish these differences. Our PCA analysis effectively captures this variability, as illustrated in Figure 2 d-f. It differentiates between cells exhibiting varying degrees of migratory behavior, including both highly and less migratory phenotypes, as well as their directionality relative to the tumor core and the persistence of their movements. Thus, we believe that our approach provides valuable insights into the distinct migratory phenotypes within the tumor microenvironment.

While our current manuscript does not provide explicit evidence linking each motility cluster to functional differences among the tumor cells, it is important to note that the state of the field supports the idea that cell dynamics can predict cell states and phenotypes. Research conducted by ourselves (Dekkers, Alieva et al., Nat Biotech, 2023) and others, such as Craiciuc et al. (Nature, 2022) and Freckmann et al. (Nat Comm, 2022) has shown that variations in cell motility patterns are indicative of underlying functional characteristics. For instance, cell morphodynamic features have been shown to reflect differences in cell types, T cell targeting states (Dekkers, Alieva et al., Nat Biotech, 2023), immune cell types (Crainiciuc et al. (Nature, 2022)), tumor metastatic potential, and drug resistance states (Freckmann et al. (Nat Comm, 2022)). In the revised manuscript, we have referenced relevant studies to underscore the biological significance of these behaviors. By doing so, we hope to clarify the potential implications of our findings and strengthen the overall narrative of our research:

In discussion: “While our current study does not provide direct functional validation of the distinct motility clusters identified, existing literature strongly supports the notion that cell dynamics can serve as a proxy for functional states and phenotypic heterogeneity. Prior work, including studies by our group[19,66] as well as Crainiciuc et al.[35] and Freckmann et al.[20], has demonstrated that variations in cell motility patterns can reflect underlying functional characteristics. Specifically, cell morpho-dynamic features have been shown to correlate with differences in cell type identity, T-cell engagement, metastatic potential, and drug resistance states. This growing body of evidence suggests that tumor cell behavior, as captured by BEHAV3D-TP, may serve as a predictive tool for deciphering functional tumor heterogeneity. Future studies integrating transcriptomic or proteomic profiling of motility-defined subpopulations could further elucidate the biological significance of these behavioral phenotypes.”

Comment #3: Using only motility to classify tumor cell behaviours in the tumor microenvironment (TME) is probably not sufficient to capture the tumor cell difference. There are also other non-tumor cell types in the TME. If the authors aim to develop a computational tool that can elucidate tumor cell behaviors in the TME, they should consider other tumor cell features, e.g., morphology, proliferation state, and tumor cell interaction with other cell types, e.g., fibroblasts and distinct immune cells.

The authors should expand the scale of tumor behavior features to classify the tumor phenotype clusters, e.g., to include tumor morphology, proliferation state, and tumor cell

| *interaction with other TME cell types.*

We believe that using dynamic features alone is sufficient to capture differences in tumor behavior, as demonstrated by our results in Figure 2. However, we appreciate the reviewer's suggestion to consider additional features, such as cell morphology, to finetune our analyses. To this end, we have adapted our pipeline to be compatible with any dynamic, morphologic or spatial features present in the data. In the revised manuscript we showcase this new addition with the analyses of two new dataset: 2D IVM data from healthy epithelial breast cells (Supplementary Fig 2) and 3D IVM data from adult gliomas (Supplementary Fig 3). These analyses identified cells with specific morphodynamic characteristics, which exhibited distinct kinetic behaviors or spatial distributions.

However, we would like to point out that not all features may provide informative insights and that a wide range of features can instead introduce biologically irrelevant noise, making interpretation more challenging. For instance, in 3D microscopy, the zaxis resolution is typically lower, which can lead to artifacts like elongation in that direction. Adding morphological features that capture this may skew the analysis. Therefore, we believe that incorporating additional features should be approached with caution. We clarify these considerations in the revised manuscript to better guide users in utilizing our computational tool effectively:

In discussion: "In addition to motility-based classification, features such as tumor cell morphology, proliferation state, and interactions with the tumor microenvironment can further refine tumor phenotyping. BEHAV3D-TP allows for the selection of diverse feature types, supporting datasets that include both dynamic, morphological and spatial parameters. However, we recognize that expanding the feature set may introduce biologically irrelevant noise, particularly in 3D microscopy data where limited z-axis resolution can lead to morphological artifacts. This highlights the potential need in the future to include unbiased feature selection strategies, such as bootstrapping methods⁶⁷, to ensure the identification of meaningful and biologically relevant parameters. Careful consideration of these aspects is key to maximizing the interpretability and predictive value of analyses performed with BEHAV3D-TP."

Comment #4: The authors have already published two papers on BEHAV3D [Alieva M et al. Nat Protoc. 2024 Jul;19(7): 2052-2084; Dekkers JF, et al. Nat Biotechnol. 2023 Jan;41(1):60-69]. Although the previous two papers used BEHAV3D to analyze T cells, the basic pipeline and computational steps are similar, in particular regarding cell segmentation and tracking. The addition of a "Heterogeneity module" based on PCA analysis does not make a significant advancement in terms of image analysis and quantification.

We want to emphasize that we have no intention of duplicating our previous publications. In this manuscript, we have consistently cited our foundational papers, where BEHAV3D was first developed for T cell migratory analysis in in vitro settings. In the introduction, we clearly state that our earlier work inspired us to adopt a similar approach for analyzing cell behavior in intravital microscopy (IVM) data, addressing the specific needs and complexities of analyzing tumor cell behaviors in the tumor microenvironment.

Importantly, our new work provides several key advancements: 1) a pipeline specifically adapted for intravital microscopy (IVM) data; 2) integration of spatial characteristics from both large-scale and small-scale phenotyping; and 3) a zero-code approach designed to empower researchers without coding skills to effectively utilize the tool. We believe that these enhancements represent meaningful progress in the analysis of cell behaviors within the tumor microenvironment which will be valuable for the IVM community. We ensure that these points are clearly articulated in the revised manuscript:

In introduction: “In line with this concept of characterizing cellular dynamic properties for cell classification, we have previously developed an analytical platform termed BEHAV3D 19,21 allowing to perform behavioral phenotyping of engineered T cells targeting cancer. While BEHAV3D was initially developed to analyze T cell migratory behavior under controlled in vitro conditions, we sought to expand its application to investigate tumor cell behaviors in IVM data, where the complexity of the TME presents distinct analytical challenges. This manuscript builds on our foundational work but represents a significant advancement by adapting the pipeline specifically for IVM datasets.”

Reviewer #3 (Recommendations for the authors):

(1) If the authors want to "democratize the analysis of heterogeneous cancer cell behaviors", they should perform image segmentation and tracking using open-source codes (e.g., Cellpose, Stardisk & 3DCellTracker) and not rely on the expensive and inaccurate ImarisTrack Module for the image analysis step of BEHAV3D.

We thank the reviewer for this recommendation and as stated above we recognize that enhancing our pipeline's compatibility with open-source options is important. To this end, we have updated our tool to support data formats generated by open-source Fiji plugins like TrackMate, MTrackJ, and ManualTracking, improving compatibility with various segmentation and tracking pipelines (https://github.com/imAIgeneDream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#data-input). In the revised manuscript, we detail this new functionality and demonstrate the operation of the BEHAV3D-TP heterogeneity module using an example dataset of glioma tumors.

Additionally, we have updated the introduction to better clarify the scope of our study (See comment #1 from Review #3) and include references to existing image processing solutions.

(2) For the 7 distinctive motility clusters, the authors should provide a more detailed analysis of the differences between them. It's unclear whether the difference in retreating, slow retreating, erratic, static, slow, slow invading, and invading correspond to functional difference of the tumor cells.

As noted in the comment above, the revised manuscript now incorporates references to relevant literature that support our understanding that behavioral differences among cells are driven by their underlying functional differences (See comment #2 from Reviewer #3). Additionally, we would like to point to Figure 2d and Supplementary Fig 4 c that provide evidence of the functional distinctions between the identified clusters.

(3) The authors should expand the scale of tumor behavior features to classify the tumor phenotype clusters, e.g., to include tumor morphology, proliferation state, and tumor cell interaction with other TME cell types.

We thank the reviewer for this valuable suggestion. In the revised manuscript, we have added the flexibility to incorporate a wide range of features, including morphological ones, and enabled users to select the specific features they wish to include in their analysis. To illustrate this functionality, we have included 2 example dataset analyzed using this approach (See comment #3 from Reviewer #3). Additionally, as indicated above we emphasize the importance of careful selection and interpretation of features, as improper choices may lead to biologically irrelevant results. This clarification is intended to ensure that users apply the tool thoughtfully and derive meaningful insights.

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